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Chapter

NEWTON'S LAWS OF MOTION- REFERENCE FRAMES

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INTRODUCTION

Every motion, either linear or rotational in a specific frame of reference is governed by certain laws. Mechanics deals with these laws, with respect to any frame, inertial or non-inertial, and formulates them in a scientific manner. We have devoted previous chapter of mathematical background to formulate these laws. Linear motion is a translational motion in a straight line. The concept of 'force' is fully explained by Newton through his laws of motion. Force is responsible for linear acceleration while torque for angular acceleration. The basic ideas of **state of rest** and **uniform motion** were studied by various philosophers. According to Aristotle (384-322 BC), a constant force has to be applied on a body so as to keep it in a motion with constant velocity. Later on, Galileo (1564-1642 AD) stated that no force is required for a body to move with uniform velocity. Newton (1642-1727) was the first philosopher who formulated the laws differentiating 'state of rest' and 'state of uniform motion'. Newton's notion of space as 'Absolute space', isotropic nature for rotational invariance and homogeneity for translational invariance led to understand the laws of motion in inertial frame of reference. Any accelerated frame of reference is a non-inertial frame. Concepts of fictitious forces like Coriolis force and its applications are studied in this chapter.

2.1 SOME IMPORTANT TERMS

Having understood the well known mathematical terms like point, straight line, scalar and vector operations under mathematical background in the previous chapter, in Physics we frequently come across three other terms: (i) particle, (ii) event and (iii) observer, as we proceed further in the development of topic 'Mechanics'. It is best, therefore, to understand at the very outset their exact connotations.

(i) Particle. A particle is ideally just a piece or a quantity of matter, having practically no linear dimensions but only a position, the measure of this quantity of matter being the mass of the particle.

Such idealised particles or point-bodies are, of course, convenient mathematical fictions and have no real existence. All that is implied is that, in our actual physical problems, if we happen to be dealing with a body whose size is very much smaller than the other quantities involved, we may to a first approximation, treat the body mathematically as a point-body or a particle. Thus, for instance, the bob of a simple pendulum, in relation to the length of the string carrying it, or a planet, in relation to its distance from the sun and the stars, may be considered to be a particle. It is not always that simple, however. For, the tiny electron, very close to a charged body is not regarded as a particle. It thus depends really on the circumstances obtaining in any given case whether or not a body is to be regarded as a particle. This will become—clearer as we proceed along.

(ii) Event. An event stands in our mathematical model for anything that occurs suddenly or instantaneously at a point in space. It thus involves both, a position and a time of occurrence.

(iii) Observer. A person or an equipment which can locate, record, measure and interpret an event is called an observer.

2.2 REST AND MOTION

We are already generally familiar with these terms from our junior classes. Here, therefore, we may simply emphasize the fact that they have a meaning only in relation to a given frame of reference (see next article). A particle is taken to be at rest or in motion in relation to the given frame of reference, according as it continues, or does not continue, to occupy the same position in that particular frame of reference.

2.3 REFERENCE FRAME

A system of coordinate axes which defines the position of a particle or an event in two or three dimensional space is called a *frame of reference*. The essential requirement of a frame of reference is that it should be rigid.

The simplest frame of reference is, of course, the familiar *Cartesian system of coordinates*, in which the position of the particle is specified by its three coordinates x, y, z along the three perpendicular axes, such that if $\vec{r}$ be the *position vector* of the particle with respect to the origin, we have

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k},$$

where, $\hat{i}, \hat{j}$ and $\hat{k}$ are the *unit vectors* along the three axes.

So that, *velocity* of the particle is given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k},$$

and its *acceleration* by $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$

The origin of the frame of reference may not always be coincident with the position of the observer. The frame of reference must, however, be at rest with respect to him.

Obviously, for complete identification of an *event* in a reference frame, *i.e.*, for determination of its exact location as well as the exact time of its occurrence, we must, in addition to the usual three spatial coordinates x, y and z , have yet another coordinate, *viz.*, that of time t , perpendicular to the former three.

A reference frame, with such four coordinates, x, y, z and t , is referred to as a **space-time frame**,—the four axes defining a *four dimensional continuum*, called **space-time**.

Now, all rest and motion being relative, the position or the state of motion of the same particle (or body) may appear to be different from different frames of reference. And, we naturally prefer to use that particular frame of reference from which its motion appears to be the simplest.

Thus, for example, a stone dropped by a person from the window of a railway carriage in uniform motion appears to him to fall *vertically* down but to a person outside the carriage, it appears to take a *parabolic path*, as indicated in Fig. 2.1.

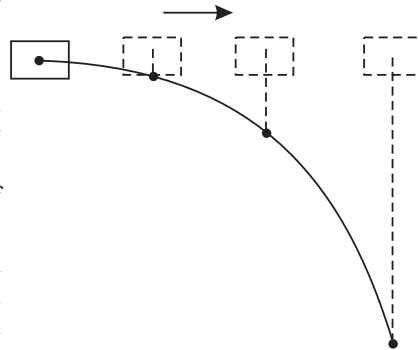


Fig. 2.1

In this case, therefore, we prefer to use a frame of reference situated in the window of the railway carriage and imagine oneself to be at its origin.

Again, if we observe the path of motion of a point on the rim of a slowly rolling wheel, we find from our frame of reference on the earth's surface that the path described by the point is a complicated, *cycloidal* one, as shown in Fig. 2.2. If, however, we have our frame of reference on the wheel itself, with its origin coincident with the centre of the wheel, and imagine oneself to be stationed there, we find that the motion of the point is a simple circular one. (See *Hint* under *Question 6*, Exercise 1).

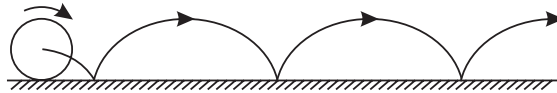


Fig. 2.2

Once again, whereas the path of a projectile appears to be parabolic from the surface of the earth, it would appear to be a straight line when viewed from another projectile (see worked example 2) and so on.

Let us now make a rapid survey of Newton's laws of motion and examine their limitations if any, in regard to the frames of reference or otherwise.

Example 1. Show that the motion of one projectile as seen from another projectile will always be a straight line motion.

Solution. Since both projectiles are subject to the same acceleration— g , there g is no relative acceleration between them. They must, therefore, appear to be descending along a straight line path relative to each other, *i.e.*, the motion of one should appear to be a straight line motion as seen from the other.

Alternatively, we may proceed as follows :

Let the trajectories of the two projectiles, projected simultaneously, be described in the x - y plane, as shown in Fig. 2.3, and let their initial velocities of projection and their angles of projection be u_1, u_2 and α_1, α_2 respectively.

Then, with the point of projection (O) as the origin if the position coordinates of points P_1 and P_2 at any given instant t be x_1, y_1 and x_2, y_2 respectively, we have $x_1 = u_1 \cos \alpha_1 t$, $x_2 = u_2 \cos \alpha_2 t$, $y_1 = u_1 \sin \alpha_1 t - \frac{1}{2} g t^2$ and $y_2 = u_2 \sin \alpha_2 t - \frac{1}{2} g t^2$. So that,

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y}{x} = \frac{u_2 \sin \alpha_2 - u_1 \sin \alpha_1}{u_2 \cos \alpha_2 - u_1 \cos \alpha_1} = m, \text{ say.}$$

whence, $y = mx$, which is the equation to a *straight line*.

The motion of one projectile, as seen from the other, is thus a straight line motion.

Example 2. To an observer, at rest on the ground, a body, thrown vertically upwards in a uniformly moving frame S , appears to describe a parabolic path. What is the path of the body as it would appear to an observer in another reference frame S' in uniform motion parallel to S , when S' has (i) an identical velocity with S , (ii) a velocity equal and opposite to that of S , and (iii) a velocity twice that of S ?

Solution. Taking the parabolic path of the body in frame S to be described in the x - y plane (where the x -axis is along the horizontal), if the x and y -components of the velocity of the body in this frame be v_x and v_y , clearly, v_x is also the velocity of frame S , along the x -axis.

In case (i), since frame S' moves with the *same* velocity as S , the x -component of the velocity of the body, as seen from S' will be, say, $v'_x = v_x - v_x = 0$ and its y -component v'_y will obviously remain the same as that in frame S , *viz.*, v_y , since neither frame has a vertical velocity.

The path of the body, therefore, will appear to an observer in frame S' to be a *vertical straight line*.

In case (ii), since frame S' moves with an equal and *opposite* velocity to that of S , the x -component of the velocity of the body, as seen from frame S' , will obviously be $v'_x = v_x - (-v_x) = 2v_x$, with the y -component, $v'_y = v_y$, as in case (i).

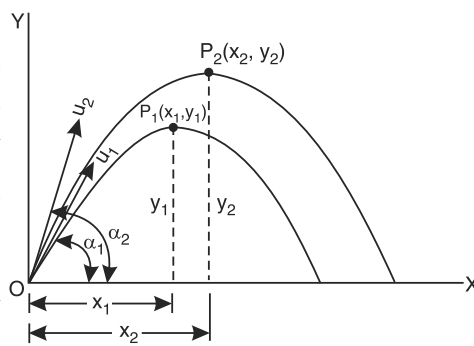


Fig. 2.3

So that, in time t , the *horizontal distance covered will be* $x = 2v_x t$ and the vertical distance covered will be $y = v_y t - \frac{1}{2} g t^2$. Or, eliminating t , we have

$$y = v_y \frac{x}{2v_x} - \frac{1}{2} g \frac{x^2}{4v_x^2}, \quad \text{Or,} \quad y = \frac{v_y}{2v_x} x - \frac{g}{8v_x^2} x^2$$

which is an equation of the form $y = ax + bx^2$, the equation to a *parabola*, indicating that the path of the body in S will, in this case, appear to the observer in S' to be a *parabola, described in the same direction* as it appears to the observer at rest on the ground. Only, the x -component of the velocity of the body ($2v_x$) now appearing to be twice as much as to the stationary observer, the latus rectum of the *parabola* will be twice that of the one seen by the observer on the ground.

In case (iii), the frame S' having twice the velocity of S , we shall have x -component of the velocity of the body, as seen from S' , i.e., $v'_x = v_x - 2v_x = -v_x$ and the y -component, $v'_y = v_y$, as before.

This is again a case similar to that of the stationary observer on the ground, with only the x -component of the velocity of the body in S now reversed. To the observer in S' , therefore, the path of the body in S will appear to be a *parabola*, similar in size and shape to that seen by the ground based observer but *described in the opposite direction*, i.e., a *parabola* which is a mirror-image of the one seen by the latter.

2.4 ABSOLUTE SPACE; TIME AND MOTION

In physics, the concept of **absolute time** and **absolute space** are hypothetical concepts closely tied to the thought of Newton. Absolute, true and mathematical time of itself, and from its own nature flows equally without regard to anything external. There is another term **duration** measures relative, apparent and common time, is same sensible and external measure of duration by means of motion which is commonly used instead of true time. Using this definition, **time** runs at the same rate for all observers in the **universe** and different measures of absolute time can be scaled by multiplying by a constant.

Thus, according to Newton, '**Absolute space**, in its own nature, without regard to anything, external, remains always similar and imovable'. Relative space is some movable dimension or measure of the absolute spaces; which our senses determine by its position to bodies.

Absolute motion is the translation of a body from one absolute place into another, and relative motion, the translation from one relative place into another.

These notions imply that absolute space and time do not depend upon physical events, but are a backdrop or stage setting within which physical phenomenon occurs. Thus, every object has an absolute state of motion relative to absolute space, so that an object must be either in a state of **absolute rest** or moving at some **absolute speed**. In modern view, the absolute space is unnecessary. Hence in classical or Newtonian mechanics, the notion of absolute space is replaced by the idea of **inertial frame of reference**; that is, a preferred **set of frames of reference** that move uniformly with respect to one another. The laws of physics transform from one inertial frame to another according to **Galilean relativity**. Newton himself recognized the role of inertial frames. Thus, the motions of bodies included in a given space are the same among themselves, whether that space is at rest or moves with uniform velocity forward in a **straight line**. As a practical matter, inertial frames often are taken as frames moving uniformly with respect to the fixed stars. The best approximation to an inertial frame is the frame of reference in the intergalactic space. According to Newton, main classical properties of space are:

(i) **Space is three dimensional**. The concept of space and time is fundamental to the study of mechanics. All objects occupy space and have a length, a breadth and a height. Space is, therefore, **three dimensional**. This is why the position of a point can be specified completely by the three coordinates (x, y, z) or (r, θ, ϕ) .

(ii) **Space is flat.** Space is flat, *i.e.*, it possess Euclidian flatness. This means that the shortest distance between any two points in space is a straight line. If we take three points in space to form a triangle, the sum of angles is equal to π . If it is a right angled triangle, then the three sides are related by Pythagoras, *i.e.*, $\text{Hypotenuse}^2 = \text{Base}^2 + \text{Altitude}^2$.

However, according to latest theory, space is **not exactly flat** but somewhat curved. The departure from flatness is very small and can be ignored in the study of classical mechanics.

(iii) **Space is homogeneous.** In other words, space is everywhere alike. Homogeneity of free space (a space in which fields and forces are absent) means translational invariance of its properties, *i.e.*, the result of an experiment is not altered due to linear displacement of the coordinate systems.

(iv) **Space is isotropic.** It means that if we consider a point O in space and we move from this point in any direction, the properties are the same *i.e.*, there is nothing to distinguish one direction from the other. In other words, there is no preferred direction in space or one direction is as good as any other direction.

Distinction between Homogeneity and Isotropy

The **homogeneity of space** means **translational invariance** of the properties of space.

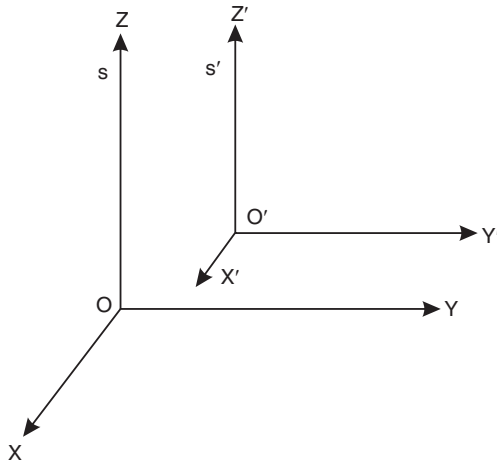


Fig. 2.4 (a) : Translational invariance.

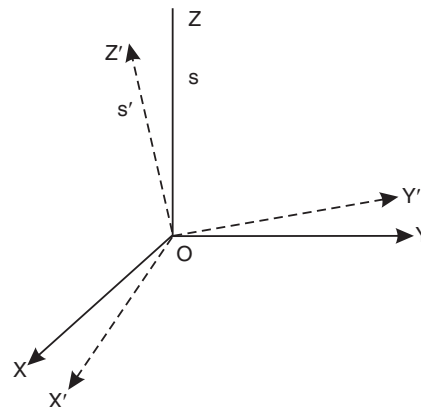


Fig. 2.4 (b) : Rotational invariance.

If we consider two systems S and S' (both are cartesian coordinate systems) displaced with respect to one another as shown in fig. 2.4 (a), then an experiment performed in system S will give exactly the same results as in system S' .

On the other hand, **isotropy** means **rotational invariance** of free space. Thus, if we have two systems S and S' rotating with respect to one another as shown in fig. 2.4 (b), then again an experiment performed in system S will give exactly the same result as in system S' .

2.5 NEWTON'S LAWS OF MOTION

It is our common experience, that a book or a duster lying on a table will continue to remain there (state of rest) until we remove it or displace it. Thus, all bodies which are initially at rest will continue to remain at rest. Similarly a body, moving with uniform velocity (state of uniform motion), will continue its uniform motion. These, state of rest or uniform motion will be continued for an infinite time, unless acted upon by an external force. This is nothing but Newton's first law of motion, Newton's three laws of motion are as under:

First law. *A body must continue in its state of rest or of uniform motion along a straight line unless acted upon by an external force.*

Second law. *The rate of change of momentum is proportional to the impressed force and takes place in the direction of the force.*

Third law. *To every action, there is an equal and opposite reaction.* Thus, if $\vec{F}_{12}$ and $\vec{F}_{21}$ be the force exerted on each other by two interacting bodies respectively, we have $\vec{F}_{12} = -\vec{F}_{21}$.

2.6 NEWTON'S FIRST LAW OF MOTION IS SIMPLY A SPECIAL CASE OF SECOND LAW

Newton's second law of motion gives a measure of a force as **the rate of change of linear momentum**. If $\vec{p}$ is the momentum of the particle, given by $\vec{p} = m\vec{v}$, where m is mass and $\vec{v}$ its velocity, then

$$\begin{aligned}\vec{F} = \frac{d\vec{p}}{dt} &= \frac{d}{dt}(m\vec{v}) \\ &= m \frac{d\vec{v}}{dt} \quad (\because m \text{ being constant}) \\ &= m\vec{a}\end{aligned}$$

So that, if $\vec{F} = 0$, then $\vec{a} = 0$ which is the first law of motion *i.e.*, if no force acts, there will be no change in velocity and hence no acceleration. Thus, **the Newton's first law of motion is simply a special case of the second law**. Hence, it is that the second law is taken to be the **real law of motion**.

2.7 LIMITATIONS OF NEWTON'S LAWS OF MOTION

The followings are the limitations:

First Limitation. The relation $\vec{F} = m\vec{a}$ (second law) would not be good in case m does not remain constant (refer article 2.6). The examples are:

(i) **A falling raindrop**, which gathers mass as it falls right from its conception within the cloud, due to more and more water vapour condensing around it. The mass of the drop, as it grows, continuously increases and therefore does not obey the condition $\vec{F} = m\vec{a}$.

(ii) **A rocket**, which continuously loses part of its mass and it moves forward, in the form of ejected burnt fuel (in succeeding stages of the rocket), and

(iii) **Particles with relativistic velocities**, The mass of a particle is given by $m = m_0 / \sqrt{1 - \frac{v^2}{c^2}}$, where m_0 is the rest mass of the particle. The mass of the particle increases as its velocity becomes appreciable fraction of the velocity of light (c).

Second Limitation. Newton's third law of motion implies that the forces exerted by the two interacting bodies over each other are equal and opposite ($\vec{F}_{12} = -\vec{F}_{21}$), provided both the forces are measured simultaneously. A simultaneous measurement of the two forces is possible for ordinary practical purposes, if the time taken by the interaction is sufficiently large compared with the time taken by light signal to travel one to the other. This means that the law ceases to hold good for particles of atomic dimensions, for which simultaneous measurement of two forces is almost impossible (refer Heisenberg uncertainty principle).

2.8 FRAMES OF REFERENCE: INERTIAL AND NON-INERTIAL

In article 2.6 and 2.7, we have discussed the Newton's laws of motion and their limitations. A *very important point about the first two laws is that they do not hold good in each and every frame of reference*. For a body at rest in one reference frame may not necessarily appear to be so in another. It may, for instance, appear to be moving in a circle in a frame of reference rotating with

respect to the first. It is only in a very special frame of reference that the two laws of motion hold good. Such a frame of reference is called a **Newtonian, inertial** or **Galilean frame of reference** (because it was *Galileo* who first enunciated the law of inertia which we refer to as Newton's first law of motion). The first law defines and the second identifies such a frame of reference to be *non-accelerating* and, therefore, also *non-rotating*. This means that in an inertial frame, the *acceleration* $\vec{a} = d^2 \vec{r} / dt^2 = 0$ (because the force applied, $\vec{F} = m \vec{a}$ is zero), or that $d^2 x / dt^2 = 0$, $d^2 y / dt^2 = 0$ and $d^2 z / dt^2 = 0$.

Inertial frame of reference. Newton's first law of motion is also known as the *law of inertia*. A reference frame e.g., a coordinate system in which Newton's first law of motion holds good, is known as an inertial frame of reference. In an inertial frame a body continues in its state of *rest* or of *uniform motion in a straight line* as long as no external force acts on it. All the Newton's laws of motion hold good in an inertial frame. All frames of reference moving with a *constant velocity* with respect to an inertial frame are also inertial frames of reference.

Consider an inertial frame of reference S and another reference frame S' (with or without their respective coordinate axes parallel to each other), such that their orientations in space remain fixed and are thus free to have only translatory motion with respect to one another. Suppose further that, with the origins O and O' at the two systems *initially coincident*, system S' has moved into the present position, shown in fig. 2.5 in time t , with a *constant velocity* $\vec{v}$ relative to S , so that $\vec{OO'} = \vec{R} = \vec{v} t$.

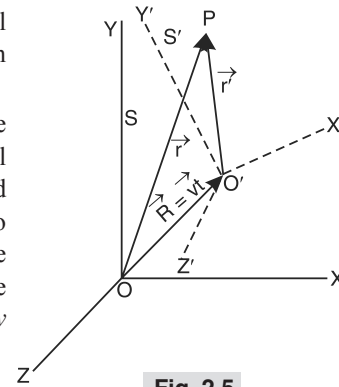


Fig. 2.5

The position vectors $\vec{r}$ and $\vec{r'}$ of a particle P , corresponding to the two frames of reference are obviously related as $\vec{r} = \vec{R} + \vec{r'}$, whence $\vec{r'} = \vec{r} - \vec{R} = \vec{r} - \vec{v} t$

So that, we have $d^2 \vec{r} / dt^2 = d^2 \vec{r'} / dt^2$ [$\vec{v}$ being constant]

Clearly, $d^2 \vec{r'} / dt^2$ is the acceleration of the particle in frame S' and $d^2 \vec{r} / dt^2$, its acceleration in the inertial frame S .

Thus, the acceleration of the particle is the *same* in the two frames of reference, indicating that if the particle be at rest in the inertial frame S , it would also appear to be at rest in frame S' . In other words, frame S' it would also be an inertial frame.

We thus arrive at a general rule that *all frames of reference, moving with a constant velocity with respect to an inertial frame, are also inertial frames of reference.*

This means, as mentioned already, that *inertial frames are non-accelerating frames or else*, a body at rest or in uniform motion in one frame may appear to have an accelerated motion in the other.

To understand more about inertial frame of reference, listen carefully the dialogue given in the following box-1.

Box-1

In a book entitled *The Evolution of Physics*, Einstein and Infeld discuss the question of the inertial system using an imaginary dialogue.

Let us interview the classical physicist and ask him some simple questions:

“What is an inertial system?”

Contd...

“It is a CS [coordinate system or what we would call a frame of reference] in which the laws of mechanics are valid. A body on which no external forces are acting moves uniformly in such a CS. This property thus enables us to distinguish an inertial CS from any other.”

“But what does it mean to say that no forces are acting on a body?”

“It simply means that the body moves uniformly in an inertial CS.”

Here we could once more put the question: “What is an inertial CS?”

But since there is little hope of obtaining an answer differing from the above, let us try to gain some concrete information by changing the question:

“Is a CS rigidly connected with the earth an inertial one?”

“No; because the laws of mechanics are not rigorously valid on the earth, due to its rotation. A CS rigidly connected with the sun can be regarded for many problems as an inertial CS; but when we speak of the rotating sun, we again understand that a CS connected with it cannot be regarded as strictly inertial.”

“Then what, concretely, is your inertial CS, and how is its state of motion to be chosen?”

“It is merely a useful fiction and I have no idea how to realize it. If I could only get far away from all material bodies and free myself from all external influences, my CS would then be inertial.”

“But what do you mean by a CS free from all external influences?”

“I mean that the CS is inertial.”

Once more we are back at our initial question!

Non-inertial frame of reference. The basic laws of physics are not changed in form in inertial frames of reference. But when the frame of reference is accelerated relative to an inertial frame, the form of basic physical laws such as Newton’s second law of motion become completely different. Such frames of reference having an accelerated motion relative to an inertial frame are called *non-inertial frames of reference*. Since a uniformly rotating frame has a centripetal acceleration it is also a non-inertial frame.

Example. A non-inertial frame is either a frame having uniform linear acceleration or a frame which is uniformly rotating.

As will be readily seen, the rotating frame of reference, referred to above, in which a body, which is at rest in an inertial frame, appears to be moving in a circle (and thus having an acceleration) is *not* an inertial frame of reference. This again implies, as mentioned earlier, that *inertial frames are also non-rotating frames*.

Further, since Newton’s laws of motion are obeyed only in inertial frames of reference, it follows as a natural consequence that, subject to the limitations mentioned under 2.7, *i.e., mass remaining constant, the relation $\vec{F} = m\vec{a}$ holds good only in inertial frames, not in non-inertial ones*.

Earth: A non-inertial frame of reference. Now, an inertial frame of reference being an unaccelerated one, the question arises as to where to actually locate it. Obviously, the earth is not a suitable place for it since, as we know, apart from going round the sun, it also spins about its own axis. This imparts to a body in the frame of reference, located on the earth, a centripetal acceleration, whose value at the equator would be $\omega^2 R = (2\pi/24 \times 60 \times 60)^2 \times 6.4 \times 10^8 = 3.4 \text{ cm/s}^2$, which is by no means negligible. Nevertheless, being quite small for most of our everyday purposes, it is quite often neglected, and *the earth taken to be a satisfactory inertial frame of reference*. Hence, also the interior of a vehicle (train, car etc.), moving smoothly, with constant velocity, on the earth’s surface, may be taken to be an equally satisfactory inertial frame of reference.

Strictly speaking, however, the earth being not quite satisfactory for the purpose, where else can we think of locating the frame of reference? Not even the sun is at rest but revolves about the centre of our galaxy, which itself moves through space. The nearest to something fixed in space are the *fixed stars*, so called because they seem to preserve their arrangement. Not that we know them to be actually fixed in their positions in space, but their velocity from this far off distance on the

earth appears to be almost negligible. *Any rigid body, therefore, situated in remote space, far from attracting matter and fixed with respect to the fixed stars, may be taken to be an inertial frame of reference.*

Now, knowing full well that, in actual practice, motion is always described in a relative frame of reference, *i.e.*, in a frame in which the position of a particle or a body is specified in relation to other material objects which may themselves be in motion relative to other objects, Newton insisted that there must exist a *fundamental or an absolute frame of reference*, which he called **absolute space**, with respect to which all motion must be measured. With the weight of his great authority behind the idea, the futile quest for this elusive absolute reference frame continued for long. But more of it later, when we deal with the special theory of relativity in the next chapter.

Here, we may simply state that for most of our purposes, the reference frame, stationary with respect to the fixed stars, is good enough as our *fixed or absolute inertial frame of reference*. All other frames of reference in uniform motion relative to it, naturally, also serve as *equivalent inertial frames*, as also any reference frame whose origin coincides with that of an inertial frame even though its coordinate axes may be inclined to those of the latter (see § 2.12).

The term *Galilean frame of reference* is used mostly in connection with *relativity*. Though here too, the earth may well be used as a Galilean frame in quite a number of cases in which its gravitational attraction is small compared with the other forces involved and its rotation is of no importance.

It only remains to be pointed out that *the concept of an inertial frame of reference is of real importance in physics for the following reasons.*

(i) *It is in respect of such a frame of reference that all our fundamental laws of physics have been formulated.*

(ii) *It is one of the basic assumptions of the special theory of relativity that the fundamental laws of physics can all be so expressed as to assume the same mathematical form in all inertial frames of reference.*

(iii) *It is endowed with the property of being isotropic with respect to mechanical and optical experiments ('isotropic' meaning same in all directions).*

2.9 GALILEAN TRANSFORMATION

An event or a physical phenomenon, observed simultaneously in two separate reference frames, will naturally have two separate sets of coordinates corresponding to the two frames of reference. Equations relating the two sets of coordinates of the event in the two systems are called **transformation equations**, because they enable the observations made in one system to be *transformed*, as it were, into those made in the other. And, if the two systems happen to be inertial or Galilean ones, the transformation is referred to as a *Galilean transformation*. Let us consider a few such transformations by way of illustration.

(a) **Transformation of position.** Let S and S' be two inertial frames of reference with observers stationed at their origins O and O' respectively, moving with a relative velocity v along the positive direction of x (Fig. 2.6). Or, what is the same thing, let S be at rest and S' move with uniform velocity v along the $+x$ direction. Needless to mention that the velocities we are considering here are all very much less than c , the velocity of light in free space, *i.e.*, $v \ll c$.

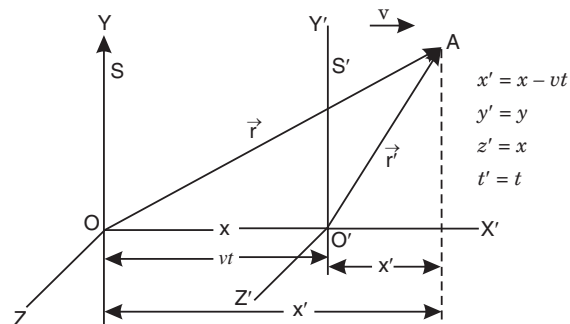


Fig. 2.6

Suppose some event occurs at A . Then, measuring time from the instant the observers were just opposite each other, *i.e.*, setting the clocks in S and S' to *zero* when origins O and O' of the two frames coincide with (or are opposite to) each other, the observer in frame S determines the position of the event by the coordinates x, y, z and the observer in S' , by the coordinates x', y', z' . Since frame S' is in uniform motion relative to S , the x -measurements in S exceed those in S' by vt , the distance covered by S' in time t in the $+x$ direction. We, therefore, have $x' = x - vt$. And, since there is no relative motion between S and S' along the axes of y and z , we have $y = y'$ and $z = z'$.

Now, if we assume time to be independent of any frame of reference, we have the four transformation equations from S to S' , *viz.*,

$x' = x - vt$	...(i)
$y' = y$	...(ii)
$z' = z$	...(iii)
$t' = t$	...(iv)

Or, we could express the transformation in a general form in terms of the position coordinates of A in the two reference frames, as explained in § 2.8, *viz.*, as

$$\vec{r}' = \vec{r} - \vec{v}t.$$

The *inverse transformation* (from S' to S) will naturally give

$$x = x' + vt, \quad y = y', \quad z = z', \quad \text{and} \quad t = t'. \quad \text{Also, } \vec{r} = \vec{r}' + \vec{v}t.$$

(b) Transformation of distance or length. Suppose there are two events occurring simultaneously at A_1 and A_2 , some distance apart, along the axis of x , such that their coordinates, as measured in the two frames of reference, are $x_1, y, z; x_2, y, z$ and $x'_1, y', z'; x'_2, y', z'$ respectively. Then, clearly,

Transformation equations for the event at A_1 (from S to S') are

$$x'_1 = x_1 - vt, \quad y' = y, \quad z' = z.$$

And transformation equations for the event at A_2 (from S to S') are

$$x'_2 = x_2 - vt, \quad y' = y, \quad z' = z.$$

Hence, *distance between the two events* is given by

$$x'_2 - x'_1 = \Delta x' = (x_2 - vt) - (x_1 - vt) = (x_2 - x_1) = \Delta x,$$

indicating that *the distance between the two events, measured by the observer in either frame of reference, is the same, irrespective of the values of v and t (which need not be known). Only, the observations must be made simultaneously in the two frames of reference.*

It will be easily seen that if Δx be the *length L of a rod*, placed along the axis of x in frame S and $\Delta x'$, its length L' along the axis of x' in frame S' , we have $L' = \Delta x' = \Delta x = L$, *i.e.*, *it is the same in either frame of reference.*

Or, we could arrive at the same result more directly by considering the end coordinates of the rod in the two frames of reference. Let these be x_1, y_1, z_1 and x_2, y_2, z_2 in frame S and x'_1, y'_1, z'_1 and x'_2, y'_2, z'_2 in frames S' . Then, clearly *length of the rod in frame S , i.e.*,

$$L = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

and its *length in frame S' , i.e.*,

$$L' = \sqrt{(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2}$$

Since, as we have seen, $x'_2 - x'_1 = x_2 - x_1$, and similarly, $y'_2 - y'_1 = y_2 - y_1$ and $z'_2 - z'_1 = z_2 - z_1$, we have

$$\boxed{L' = L}$$

i.e., *the length of the rod is the same in either frame by reference.*

Now, when a particular property is characterised by the same number in the two frames of reference, it is said to be '*invariant to the transformation*'. So that, we may say that **a distance or a length is invariant to Galilean transformation.**

(c) Transformation of velocity. Let us now consider the velocity measurements u and u' as made by the two observers in their respective frames of reference. For this, we consider a displacement in the $+x$ direction, say, and the time interval for it. Thus, if x_1, t and x_2, t be the coordinates of the initial and final events characterising the displacement in frame S and x'_1, t'_1 and x'_2, t'_2 , the corresponding coordinates in frame S' , we have

$$\text{velocity relative to frame } S, \text{ given by } u = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Or, since $x_1 = x'_1 + vt'_1$, $x_2 = x'_2 + vt'_2$, $t_1 = t'_1$ and $t_2 = t'_2$ we have

$$u = \frac{(x'_2 + vt'_2) - (x'_1 + vt'_1)}{t'_2 - t'_1} = \frac{x'_2 - x'_1}{t'_2 - t'_1} + v \frac{(t'_2 - t'_1)}{(t'_2 - t'_1)}$$

$$\text{Or } u = \frac{\Delta x}{\Delta t'} + v = u' + v. \quad \text{Or, } u' = u - v. \quad \dots (vi)$$

Again, we could arrive at the same result from relation (v) above, viz, $\vec{r}' = \vec{r} - \vec{v}t$, differentiating which, we have

$$\frac{d\vec{r}'}{dt} = \frac{d\vec{r}}{dt} - \vec{v}. \quad \text{Or, } \boxed{\vec{u}' = \vec{u} - \vec{v}} \quad \dots (vii)$$

indicating that the velocities measured by the observers in the two frames of reference are not the same. We, therefore, say that **velocity is not invariant to Galilean transformation.**

The *inverse transformation* (from frame S' to S) is, obviously, given by

$$\vec{u} = \vec{u}' + \vec{v}. \quad \dots (viii)$$

Here, relation (vii) or (viii) is often referred to as the **Galilean law of addition of velocities.**

(d) Transformation of acceleration. Considering accelerations $\vec{a}$ and $\vec{a}'$, as measured by the observers in their respective frames of reference, we have $\vec{a} = d\vec{u}/dt$ and $\vec{a}' = d\vec{u}'/dt$.

Since, as just seen under (c), above, $\vec{u} = \vec{u}' + \vec{v}$, we have

$$\frac{d\vec{u}}{dt} = \frac{d\vec{u}'}{dt} + 0. \quad (\vec{v} \text{ being a constant.})$$

Or,

$$\boxed{\vec{a} = \vec{a}'}$$

indicating that the accelerations, as measured by the two observers in the two frames of reference respectively, are the same. **Acceleration is thus invariant to Galilean transformation.**

Example 3. A moving particle has co-ordinates $(6t + 3)$, $8t$, 5m in frame S at any time t . The frame S' is moving relative to S with a velocity $3\hat{i} + 4\hat{j}$ m/s. Find the co-ordinates and velocity of the particle in frame S .

Solution. Galelian transformation equations for space co-ordinates are

$$x' = x - v_x t \quad y' = y - v_y t \quad z' = z - v_z t$$

$$\text{Now } x = 6t + 3 \quad y = 8t \quad z = 5; \quad v_x = 3 \quad v_y = 4 \quad v_z = 0$$

$$\therefore x' = x - v_x t = 6t + 3 - 3t = 3t + 3$$

$$y' = y - v_y t = 8t - 4t = 4t; \quad z' = z - v_z t = 5 - 0 = 5$$

$\therefore$ Co-ordinates of the particle in S' are $3t + 3$, $4t$, 5 .

$$\text{Velocity } x' = 3t + 3 \quad \therefore \frac{dx'}{dt'} = \frac{dx'}{dt} \cdot \frac{dt}{dt'} = \frac{dx'}{dt} \quad \dots \left[\begin{array}{l} \because t' = t \\ dt' = dt \end{array} \right]$$

$$\text{Hence } \frac{dx'}{dt} = 3 \text{ or } \vec{u}'_x = 3\hat{i}$$

$$\text{Similarly } \frac{dy'}{dt'} = \frac{dy'}{dt} = 4 \quad \text{or} \quad \vec{u}'_y = 4\hat{j} \quad \frac{dz'}{dt'} = \frac{dz'}{dt} = 0$$

$$\therefore \text{Velocity in frame } S', \quad \vec{u}' = \vec{u}'_x + \vec{u}'_y + \vec{u}'_z = 3\hat{i} + 4\hat{j}$$

Example 4. The position vector of a point in the frame S' moving with constant velocity of 10 cm/sec along X -axis is given by (11, 9, 8) cm. Calculate the position with respect to the frame S if the two frames were coincident only $\frac{1}{2}$ sec earlier. (P.U. 2001, 2000)

Solution. As the position co-ordinates are given in the frame S' which is moving with constant velocity of $v = 10 \text{ cm s}^{-1}$ along $+X$ -axis, the position co-ordinates in the frame S are given by inverse Galelian transformations

$$x = x' + vt; \quad y = y'; \quad z = z'; \quad t = t'$$

$$\text{Here} \quad x' = 11 \text{ cm} \quad y' = 9 \text{ cm} \quad z' = 8 \text{ cm} \quad t' = \frac{1}{2} \text{ sec.}$$

$$\therefore \quad x = x' + vt = 11 + 10 \times \frac{1}{2} = 16 \text{ cm}; \quad y' = 9 \text{ cm}; \quad z' = 8 \text{ cm}$$

$\therefore$ The position co-ordinates of the point in the frames S are (16, 9, 8) cm.

Example 5. A frame S' is moving with velocity $5\hat{i} + 7\hat{j} \text{ m/s}$ relative to an inertial frame S . A particle is moving with velocity $(t+5)\hat{i} + 9\hat{j} \text{ m/s}$ with respect to S . Find the acceleration of the particle in the frame S' .

Solution. Velocity of the particle in frame S , $\vec{u} = (t+5)\hat{i} + 9\hat{j} \text{ ms}^{-1}$

Velocity of frame S' relative to S , $\vec{v} = 5\hat{i} + 7\hat{j}$

According to Galelian transformation equations for velocity,

Velocity of the particle with respect to frame S' ,

$$\vec{u}' = \vec{u} - \vec{v} = (t+5)\hat{i} + 9\hat{j} - (5\hat{i} + 7\hat{j}) = t\hat{i} + 2\hat{j} \text{ ms}^{-1}$$

$$\therefore \text{ Acceleration of the particle in the frame } S', \quad \vec{a}' = \frac{d\vec{u}'}{dt} = \frac{d}{dt} \quad \dots [\because dt' = dt]$$

$$= \frac{d}{dt} [t\hat{i} + 2\hat{j}] = \hat{i} \quad \therefore \vec{a}' = \hat{i} \text{ ms}^{-2}$$

Example 6. The pilot of an airplane wishes to reach a point 200 km east of his present position. A wind blows 30 km/hr from the north west. Calculate his vector velocity with respect to the moving air mass if his schedule requires him to arrive at his destination in 40 minutes.

(Take $\sqrt{2} = 1.4$).

Solution. Let O be the position of the pilot at the given instant (Fig. 2.7). Since he reaches the desired point 200 km due east in 40 min., his velocity in this direction is $200 \times 60/40 = 300 \text{ km/hr}$.

The problem thus reduces to this, that 300 km/hr is the resultant of the wind velocity v_w and the pilot's own velocity v_p , say. By the Triangle law of vectors,

therefore, $\vec{OA}$ represents the wind's velocity vector $\vec{v}_w$, $\vec{AP}$, the velocity vector $\vec{v}_p$ of the pilot and $\vec{OP}$

the resultant velocity vector $\vec{R}$. So, that, $\vec{R} = \vec{v}_w + \vec{v}_p$, whence,

$$\vec{v}_p = \vec{R} - \vec{v}_w.$$

Now, $\vec{v}_w = 30 \sin 45^\circ \hat{i} - 30 \cos 45^\circ \hat{j} = 15\sqrt{2}\hat{i} - 15\sqrt{2}\hat{j}$ and $R = 300\hat{i}$, where $\hat{i}$ and $\hat{j}$ are the unit vectors along the east and the north directions respectively.

We, therefore, have $\vec{v}_p = 300\hat{i} - (15\sqrt{2}\hat{i} - 15\sqrt{2}\hat{j}) = 300\hat{i} - 21(\hat{i} - \hat{j})$.

The vector velocity of the pilot with respect to the moving mass of air is thus $300\hat{i} - 21(\hat{i} - \hat{j})$.

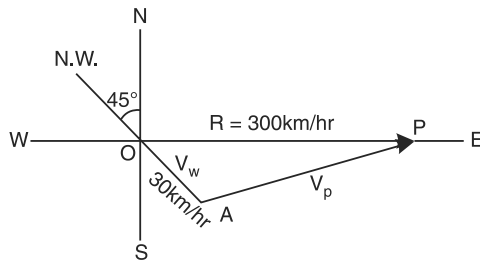


Fig. 2.7

Example 7. A reference frame S' is moving with a constant velocity $\vec{v}$ with respect to an inertial frame S . If at $t = 0$, the position vector of O' (the origin of the moving frame S') with respect to O (the origin of the inertial frame S) be $\vec{R}$, show that the position vectors $\vec{r}$ and $\vec{r}'$ of a point P with respect to frames S and S' respectively are related by the equation $\vec{r} = \vec{R} + \vec{v}t + \vec{r}'$.

Solution. Since the reference frame S' is moving with a constant velocity with respect to the inertial frame S , it is also an inertial frame. The problem thus reduces to one of Galilean transformation of the position vector of a point in one Galilean frame of reference to that in the other.

Since, initially, at time $t = 0$, the position vector of the origin O' of the moving frame S' with respect to the origin O of the inertial frame S , is $\vec{R}$, it would be $(\vec{R} + \vec{v}t)$ in time t , when the former would have covered an additional distance $\vec{v}t$. So that, vector $\vec{OO'}$ would be equal to $\vec{R} + \vec{v}t$.

Referring to Fig. 2.3 which depicts the whole situation except that $\vec{OO'}$ now represents $(\vec{R} + \vec{v}t)$ instead of $\vec{R}$ (because, in the case shown, O' coincided with O at $t = 0$), let us consider to position vectors $\vec{r}$ and $\vec{r}'$ of a point P with respect to the origins of the two frames of reference. It is clear at once that vector $\vec{OP} = \text{vector } \vec{OO'} + \text{vector } \vec{O'P}$,

$$\text{i.e.,} \quad \vec{r} = \vec{R} + \vec{v}t + \vec{r}'.$$

Example 8. The velocity vector of an object in uniform circular motion with a speed of 50 cm/s changes direction by 30° in 2 seconds. Calculate (i) the change in velocity, (ii) the average acceleration, and (iii) the centripetal acceleration. How do you explain the fact that (ii) and (iii) have identical values?

Solution. (i) Let the object be moving along the circle shown in Fig. 2.8 and, taking the coordinate axes OX and OY , with the centre (O) of the circle as the origin, let P be its position at a given instant and P' , its position after 2s when the velocity vector has changed direction by an angle θ . Then, the velocities of the object at P and P' are clearly $\vec{v}_1$ and $\vec{v}_2$, tangential to OP and OP' respectively, as shown.

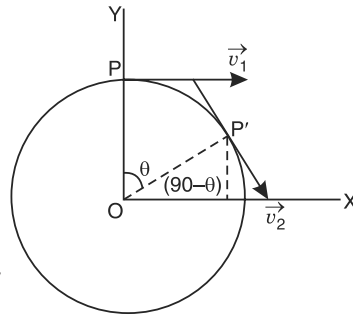


Fig. 2.8

Since only the direction of the velocity changes, its magnitude remaining the same, we have $|\vec{v}_1| = |\vec{v}_2| = v$, say. So that, expressing $\vec{v}_1$ and $\vec{v}_2$ in terms of their components along the axes of x and y , we have

$$\vec{v}_1 = v\hat{i}, \text{ the component along the } y\text{-axis being zero.}$$

$$\text{and } \vec{v}_2 = -v\sin\theta\hat{j} + v\cos\theta\hat{i} = v\cos 30^\circ\hat{i} - v\sin 30^\circ\hat{j}$$

Obviously, therefore, change in velocity, $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$

$$= v\cos 30^\circ\hat{i} - v\sin 30^\circ\hat{j} - v\hat{i}.$$

$$\text{Or, } \Delta\vec{v} = v(\cos 30^\circ - 1)\hat{i} - v\sin 30^\circ\hat{j} = v\left(\frac{\sqrt{3}}{2} - 1\right)\hat{i} + \frac{1}{2}v\hat{j}.$$

$$\text{Or, since } v = 50 \text{ cm/s, we have } \Delta\vec{v} = 50(0.134)\hat{i} - 25\hat{j} = -6.7\hat{i} - 25\hat{j}$$

$$\therefore \text{magnitude of the change in velocity} = \sqrt{(6.7)^2 + (25)^2} = 25.92 \text{ or } 26 \text{ cm/s.}$$

$$\begin{aligned} \text{(ii) Clearly, average acceleration} &= \frac{\text{Change in velocity}}{\text{time}}. \quad \text{Or, } \vec{a} = \frac{\Delta\vec{v}}{\Delta t} \\ &= \frac{-6.7\hat{i} - 25\hat{j}}{2} = -3.35\hat{i} - 12.5\hat{j}. \end{aligned}$$

$\therefore$ magnitude of the average acceleration

$$= \sqrt{(3.35)^2 + (12.5)^2} = 12.94 \quad \text{or} \quad 13.0 \text{ cm/s}.$$

(iii) Finally the centripetal acceleration of the object $= v^2/r$,

where r is the radius of its circular path.

Now, $v = r\omega$ and $\therefore r = v/\omega$, where $v = 50 \text{ cm/s}$ (given) and to determine ω , we note that the object describes an angle of 30° or $30 \times \pi/180 = \pi/6$ radians in 2 seconds. It will, therefore, describe an angle of $\pi/12$ radian in 1 second i.e., $\omega = \pi/12$. So that,

$$r = 50 / \frac{\pi}{12} = 600 / \pi = 191 \text{ cm}.$$

$$\therefore \text{centripetal acceleration of the object} = \frac{v^2}{r} = \frac{50 \times 50}{191} = 13.09 \text{ or } 13.0 \text{ cm/s}^2.$$

The reason why the values of (ii) average acceleration and (iii) centripetal acceleration agree is that although average acceleration means the average of the total acceleration consisting of (a) radial and (b) angular accelerations, here, in this case, the motion being a *uniform circular* one; the question of angular acceleration does not arise. So that, the only acceleration (or the total acceleration) is the centripetal acceleration.

Example 9. The earth, revolving around its own axis, is not, strictly speaking, an inertial frame of reference. If, however, we take it to be so, what would be the error involved in 1 second, in the position of a particle close to its surface? (Radius of the earth $= 6.4 \times 10^8 \text{ cm}$).

Solution. Obviously, due to rotation about its own axis, a particle on, or close to, the surface of the earth, experiences a centripetal acceleration equal to $\omega^2 R$, where ω is the angular velocity of the earth $= 2\pi/24 \times 60 \times 60$ radian/s and R , its radius $= 6.4 \times 10^8 \text{ cm}$ (given).

$$\therefore \text{centripetal acceleration, } a = \left(\frac{2\pi}{24 \times 60 \times 60} \right)^2 \times 6.4 \times 10^8 \text{ cm/s}^2.$$

$$= 3.385 \quad \text{or} \quad 3.4 \text{ cm/s}^2.$$

Thus, the acceleration of a particle, measured in the earth's frame of reference, will differ by 3.4 cm/s^2 from that measured in a truly inertial frame of reference, and the position of a particle, as measured in the latter, will be altered by a distance $\frac{1}{2}at^2$ in t seconds or by $\frac{1}{2}a \times 3.4$ or 1.7 cm in 1 second, when measured in the former.

The error involved in 1s in the position of the particle, as measured in the earth's frame of reference, would thus be 1.7 cm .

Example 10. The initial positions of two particles are $x_1 = 10 \text{ cm}$, $y = 0$ and $x_2 = 0$, $y = 10 \text{ cm}$ respectively. If the velocity of the first particle be $-5 \times 10^5 \hat{i} \text{ m/s}$ calculate that of the second at the instant that the two just collide with each other. What is then the relative velocity of the first particle with respect to the second?

Solution. Suppose the two particles collide after time t second. Then clearly, position vector of the first particle after time t , i.e., $\vec{r}_1(t) = \vec{r}_1 + \vec{v}_1 t = 10\hat{i} + (-5 \times 10^5 \hat{i})t$

$$= 10\hat{i} - 5 \times 10^5 \hat{i}t.$$

And position vector of the second particle after time t , i.e., $\vec{r}_2(t) = \vec{r}_2 + \vec{v}_2 t = 10\hat{j} + \vec{v}_2 t$.

In order that the two particles may collide, we must, obviously, have

$$\vec{r}_1(t) = \vec{r}_2(t).$$

$$\text{Or, } \vec{r}_1(t) - \vec{r}_2(t) = 0, \text{ i.e., } (10\hat{i} - 5 \times 10^5 \hat{i}) - 10\hat{j} - \vec{v}_2 t = 0.$$

$$\text{Or, } (10 - 5 \times 10^5 t)\hat{i} - 10\hat{j} - \vec{v}_2 t = 0, \text{ whence, } \vec{v}_2 t = (10 - 5 \times 10^5 t)\hat{i} - 10\hat{j}.$$

Since the second particle has only a velocity along the y -axis, the x -component of its velocity is zero, i.e., $10 - 5 \times 10^5 t = 0$, whence,

$$t = 10/5 \times 10^5 = 2 \times 10^{-5} \text{ s},$$

$$\text{So that, } \vec{v}_2 t = -10\hat{j}. \text{ Or, } \vec{v}_2 = -\frac{10}{t}\hat{j} = -\frac{10}{2 \times 10^{-5}}\hat{j} = -5 \times 10^5 \hat{j} \text{ cm/s.}$$

The velocity of the second particle at the time of collision is thus $-5 \times 10^5 \hat{j}$ cm/s.

And, the relative velocity of the first particle with respect to the second is, obviously,

$$\vec{v}_r = \vec{v}_1 - \vec{v}_2 = -5 \times 10^5 \hat{i} + 5 \times 10^5 \hat{j} = 5 \times 10^5 (\hat{j} - \hat{i}) \text{ cm/s.}$$

2.10 FRAMES OF REFERENCE WITH LINEAR ACCELERATION

In Fig. 2.5 let frame S' be moving with a linear acceleration $\vec{a}_0$ with respect to the *inertial frame* S . Then, particle A or, in fact, any other particle at rest with respect to frame S will clearly appear to be moving with an acceleration $-\vec{a}_0$ with respect to frame S' and, therefore, a particle having an acceleration $\vec{a}$ with respect to the inertial frame S will appear to have an acceleration $\vec{a}' = \vec{a} - \vec{a}_0$ in frame S' . So that, if m be the mass of the particle (assumed to remain unaltered in either frame), we have

observed force on the particle in frame S' given by

$$\vec{F} = m\vec{a}' = m(\vec{a} - \vec{a}_0) = m\vec{a} - m\vec{a}_0.$$

where $m\vec{a} = \vec{F}$ is clearly the force on the particle in the inertial frame S .

We, therefore, have $\vec{F}' = \vec{F} - m\vec{a}_0$

Or, putting $m\vec{a}_0 = \vec{F}_0$ we have $\vec{F}' = \vec{F} - \vec{F}_0$

And if $\vec{F} = 0$, $\vec{F}' = -\vec{F}_0$

Thus, even when no force is acting on the particle in frame S , a force $\vec{F}_0 = m\vec{a}_0$ appears to be acting on the particle in frame S' which is, therefore, a **non-inertial frame**. (see also § 2.14)

Incidentally, the relation $\vec{a}' = \vec{a} - \vec{a}_0$ or $\vec{a} = \vec{a}' + \vec{a}_0$ is called the **Galilean law of composition of accelerations**.

2.11 CLASSICAL RELATIVITY—GALILEAN INVARIANCE

In one of the corollaries to his laws of motion, Newton says : '*the motions of bodies included in a given space are the same among themselves whether that space is at rest or moves uniformly forward in a straight line*'. This clearly implies that if we are drifting along at a uniform speed in a closed space ship, *all the phenomena observed, and all the experiments performed inside the ship, will appear to be the same as if the ship were not in motion*. This is the concept of *Classical or Newtonian relativity*, usually referred to as *Galilean invariance*. It clearly means that *the fundamental physical laws and principles are all invariant* to Galilean transformation*. Or, in plain and simple language, *the fundamental physical laws and principles are identical in all inertial frames of reference*. Let us see if this is so.

(i) **Newton's law of motion.** Considering the second law of motion which in the *real* law of motion and includes also the first law, let F be the force acting on a mass m in frame S (Fig. 2.5). Then, we have $F = ma$, where a is the acceleration produced in the mass. *Assuming the mass to be independent*

of velocity, if F' be the force acting on it in frame S' , we have $\vec{F} = m\vec{a}$, where $\vec{a}'$ is the acceleration produced in it in this frame. Since, as we know, $\vec{a} = \vec{a}'$ in the two inertial frames, we have

$$\vec{F} = \vec{F}', \quad \dots (i)$$

indicating that *the law is invariant to Galilean transformation*.

* When something does not change, one says it is invariant. Covariance means no change in form, usually used for equations.

(ii) The law of conservation of momentum. The momentum of a body or a particle of mass m and velocity $\vec{v}$ is as we know, given by $\vec{p} = m\vec{v}$. And, the law of conservation of momentum merely states that if a number of particles collide against each other, their total momentum after collision is the same as their total momentum before collision, irrespective of the collision being elastic or inelastic. Only, in the particles before collision appears as their kinetic energy after collision and, in the case of an inelastic collision, part of the kinetic energy particles before collision appears in the form of heat (or some such sort of internal excitation energy) after the collision and only the balance as kinetic energy of the particles.

Thus if we consider only two particles of masses m_1 and m_2 in frame S (Fig. 2.5), moving with velocities $\vec{u}_1$ and $\vec{u}_2$ respectively before collision and with velocities $\vec{v}_1$ and $\vec{v}_2$ after collision, the law of conservation of momentum demands that

$$m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2. \quad \dots (ii)$$

Taking the masses to remain unaltered in the two inertial frames, if the law of conservation of momentum is to be invariant to Galilean transformation, it should assume, in frame S' , the form

$$m_1\vec{u}'_1 + m_2\vec{u}'_2 = m_1\vec{v}'_1 + m_2\vec{v}'_2. \quad \dots (iii)$$

where $\vec{u}'_1, \vec{u}'_2$ and $\vec{v}'_1, \vec{v}'_2$ are the respective velocities of the particles before and after collision in frame S' .

This is actually found to be the case. For, as we know, [§ 2.9 (c)].

$\vec{u}_1 = \vec{u}'_1 + \vec{v}$, $\vec{u}_2 = \vec{u}'_2 + \vec{v}$, $\vec{v}_1 = \vec{v}'_1 + \vec{v}$, and $\vec{v}_2 = \vec{v}'_2 + \vec{v}$, where $\vec{v}$ is the relative velocity of frame S' with respect to frame S .

Substituting these values in relation (ii) above, we have

$$m_1(\vec{u}'_1 + \vec{v}) + m_2(\vec{u}'_2 + \vec{v}) = m_1(\vec{v}'_1 + \vec{v}) + m_2(\vec{v}'_2 + \vec{v}).$$

$$\text{Or, } m_1\vec{u}'_1 + m_2\vec{u}'_2 + \vec{v}(m_1 + m_2) = m_1\vec{v}'_1 + m_2\vec{v}'_2 + \vec{v}(m_1 + m_2).$$

$$\text{Or, } m_1\vec{u}'_1 + m_2\vec{u}'_2 = m_1\vec{v}'_1 + m_2\vec{v}'_2. \quad \dots (iv)$$

showing that the law of conservation of momentum is also invariant to Galilean transformation. i.e., holds good in all inertial frames of reference.

It may be carefully noted that although the values of momenta and changes of momentum of the individual particles are different in the two frames of reference, the total momentum of the two is conserved in either frame. Thus, whereas momentum, by itself, is not invariant to Galilean transformation, the principle of conservation is.

(iii) The law of conservation of energy. Again, consider a collision between two particles in the inertial frame S (Fig. 2.5). Taking their respective masses to be m_1 and m_2 and their respective velocities to be $\vec{u}_1, \vec{u}_2$ and $\vec{v}_1, \vec{v}_2$ before and after collision, we have, in accordance with the law of conservation of energy,

$$\frac{1}{2}m_1\vec{u}_1^2 + \frac{1}{2}m_2\vec{u}_2^2 = \frac{1}{2}m_1\vec{v}_1^2 + \frac{1}{2}m_2\vec{v}_2^2 + E, \quad \dots (v)$$

where, E is part of the energy of the particles (before the collision) appearing in some other form, like heat etc.

Let the velocities of the two masses in frame S' before and after collision be $\vec{u}'_1, \vec{u}'_2$ and $\vec{v}'_1, \vec{v}'_2$ respectively. Then, with the values of the masses (m_1 and m_2) remaining unaltered and with E remaining the same (as established by actual experiment), the law of conservation of energy must assume the form

$$\frac{1}{2}m_1\vec{u}'_1{}^2 + \frac{1}{2}m_2\vec{u}'_2{}^2 = \frac{1}{2}m_1\vec{v}'_1{}^2 + \frac{1}{2}m_2\vec{v}'_2{}^2 + E, \quad \dots (vi)$$

if it is to be invariant to Galilean transformation.

That this is actually so may be easily seen by substituting in relation (v) above, the values of $\vec{u}_1$, $\vec{u}_2$, $\vec{v}_1$ and $\vec{v}_2$ obtained by Galilean transformation [§ 2.9 (c)], as in case (ii) above, when we have

$$\frac{1}{2} m_1 (\vec{u}_1^2 + \vec{v}^2) + \frac{1}{2} m_2 (\vec{u}_2^2 + \vec{v}^2) = \frac{1}{2} m_1 (\vec{v}_1^2 + \vec{v}^2) + \frac{1}{2} m_2 (\vec{v}_2^2 + \vec{v}^2) + E,$$

where $\vec{v}$, as we know, is the relative velocity of frame S' with respect to S .

$$\begin{aligned} \text{Or, } \frac{1}{2} m_1 \vec{u}_1^2 + \frac{1}{2} m_2 \vec{u}_2^2 + \vec{v} (m_1 \vec{u}_1 + m_2 \vec{u}_2) \\ = \frac{1}{2} m_1 (\vec{v}_1^2 + \vec{v}^2 + 2 \vec{v}_1 \vec{v}) + \frac{1}{2} m_2 (\vec{v}_2^2 + \vec{v}^2 + 2 \vec{v}_2 \vec{v}) + E. \end{aligned}$$

$$\begin{aligned} \text{Or, } \frac{1}{2} m_1 \vec{u}_1^2 + \frac{1}{2} m_2 \vec{u}_2^2 + \vec{v} (m_1 \vec{u}_1 + m_2 \vec{u}_2) \\ = \frac{1}{2} m_1 \vec{v}_1^2 + \frac{1}{2} m_2 \vec{v}_2^2 + v (m_1 \vec{v}_1 + m_2 \vec{v}_2) + E. \end{aligned}$$

Since momentum remains conserved in frame S' , we have

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2.$$

$$\text{And, therefore, } \frac{1}{2} m_1 \vec{u}_1^2 + \frac{1}{2} m_2 \vec{u}_2^2 = \frac{1}{2} m_1 \vec{v}_1^2 + \frac{1}{2} m_2 \vec{v}_2^2 + E,$$

showing that energy is also conserved in frame S' .

In other words, *the law of conservation of energy (like that of conservation of momentum) is invariant to Galilean transformation.*

Now, simply because any two reference frames, moving with constant relative velocity, can be connected by means of a Galilean transformation, we should not rush to the conclusion that the fundamental laws of Physics are invariant or identical in all reference frames that can be so connected. There is a snag here, namely, we have assumed both *mass* and *time* to be invariant to Galilean transformation. There is little or no justification for it, as we shall see later when we are faced with invariance of the velocity of light (c) in free space. More correctly, the statement should be that *the fundamental physical laws are invariant or identical in all reference frames that can be connected, not by Galilean, but by Lorentz transformations* (§ 3.6) which takes due account of the invariance of the value of c . For non-relativistic velocities (*i.e.*, for velocities very much less than c) Lorentz transformations too reduce to Galilean ones.

2.12 TRANSFORMATION EQUATIONS FOR A FRAME OF REFERENCE INCLINED TO AN INERTIAL FRAME

Let S be an inertial or a Galilean frame of reference and S' , a frame of reference with its origin coinciding with that of S but with its coordinate axes inclined to those of the latter, as shown in Fig. 2.9.

Consider a particle at P , whose position vector is $\vec{r}$ with respect to the origin of either frame of reference. Its coordinates are obviously x, y, z in frame S and x', y', z' in frame S' .

We have $x' = \text{sum of the components of } x, y, z \text{ along the axis } OX', \text{ i.e., } = x \cos XOX' + y \cos YOX' + z \cos ZOX'.$

Or, representing $\cos XOX'$ by $\cos (X, X')$ or $C_{xx'}$, $\cos YOX'$ by $\cos (Y, X')$ or $C_{yx'}$ and $\cos ZOX'$ by $\cos (Z, X')$ or $C_{zx'}$, we have

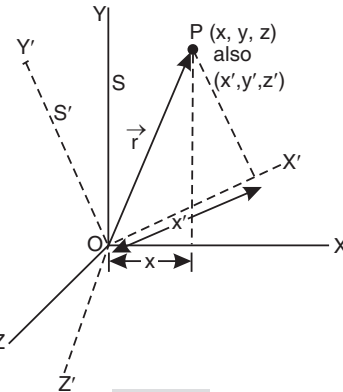


Fig. 2.9

$$x' = xC_{xx}' + yC_{yx}' + zC_{zx}' \quad \dots(i)$$

$$\text{Similarly, } y' = xC_{xy}' + yC_{yy}' + zC_{zy}' \quad \dots(ii)$$

$$\text{and } z' = xC_{xz}' + yC_{yz}' + zC_{zz}' \quad \dots(iii)$$

These are clearly the *transformation equations* from frame S to S' . Differentiating expression (i) twice with respect to t , we have

$$\frac{d^2x'}{dt^2} = \frac{d^2x}{dt^2} C_{xx}' + \frac{d^2y}{dt^2} C_{yx}' + \frac{d^2z}{dt^2} C_{zx}' \quad \dots(iv)$$

Since S is an *inertial frame*, and no force is acting on the particle at P , we have $d^2x/dt^2 = 0$, also $d^2y/dt^2 = 0$ and $d^2z/dt^2 = 0$.

Substituting these values in relation (iv), therefore, we have

$$d^2x'/dt^2 = 0.$$

In exactly the same manner we can show that

$$d^2y'/dt^2 = 0 \quad \text{and} \quad d^2z'/dt^2 = 0.$$

Obviously, then, **the reference frame S' is also an inertial frame** and hence the transformation equations (i), (ii) and (iii) above are *Galilean transformation equations from the inertial or Galilean frame S to the inertial or Galilean frame S'* .

The *inverse Galilean transformation equations* (from S' to S) will obviously be

$$x = x'C_{xx}' + y'C_{yx}' + z'C_{zx}', \quad y = x'C_{xy}' + y'C_{yy}' + z'C_{zy}'$$

$$\text{and} \quad z = x'C_{xz}' + y'C_{yz}' + z'C_{zz}'.$$

2.13 TRANSFORMATION EQUATIONS FOR A ROTATING FRAME OF REFERENCE

Consider an inertial frame of reference S and another reference frame S' whose origin and coordinate axes coincide with those of S .

Now, with the origins of the two frames still coinciding, let S' start rotating with a uniform angular velocity ω about the common axis of z , so that in time t , the axes OX' and OY' of S' have turned through angle ωt each with respect to axes OX and OY of S , as shown in Fig. 2.10.

As in article 2.12 above, we have to determine the relation between the coordinates, x, z and x', y', z' of a particle P in the two frames of reference respectively, its position vector with respect to the origin being the same $\vec{r}$ in either frame.

Proceeding exactly as in article 2.12, we have

$$x' = xC_{xx}' + yC_{yx}' + zC_{zx}', \quad y' = xC_{xy}' + yC_{yy}' + zC_{zy}' \text{ and, of course, } z' = z.$$

Since $C_{xx}' = \cos XOX' = \cos \omega t$, $C_{yx}' = \cos YOY' = \cos (\pi/2 - \omega t) = \sin \omega t$ and $C_{zx}' = \cos XOZ' = \cos (\pi/2) = 0$, we have

$$x' = x \cos \omega t + y \sin \omega t, \quad \dots(v)$$

Again, since $C_{xy}' = \cos XOY' = \cos (\pi/2 + \omega t) = -\sin \omega t$,

$$C_{yy}' = \cos YOY' = \cos \omega t \text{ and } C_{zy}' = \cos ZOY' = \cos (\pi/2) = 0,$$

$$\text{we have } y' = -x \sin \omega t + y \cos \omega t, \quad \dots(vi)$$

$$\text{and, as pointed out earlier } z' = z. \quad \dots(vii)$$

Relations (v), (vi) and (vii) are, therefore, the *transformation equations* in the case of the frame of reference S' , rotating with a uniform angular velocity ω relative to the inertial frame S .

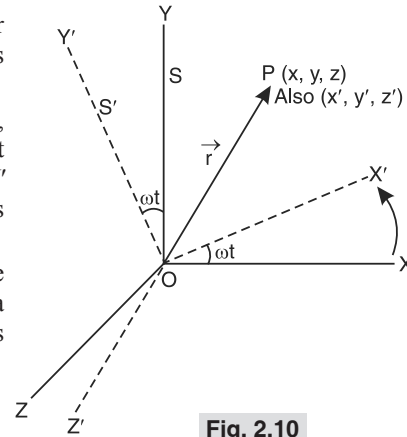


Fig. 2.10

The inverse transformation equations (from S' to S) will obviously be

$$x = x' \cos \omega t - y' \sin \omega t, \quad y = x' \sin \omega t + y' \cos \omega t \quad \text{and} \quad z = z'.$$

Now, frame S is an inertial frame, and so force is applied to particle P . We, therefore, have

$$\frac{d^2 x}{dt^2} = 0, \quad \frac{d^2 y}{dt^2} = 0 \quad \text{and} \quad \frac{d^2 z}{dt^2} = 0.$$

On the other hand, differentiating expressions (v), (vi) and (vii) for x' , y' and z' respectively with respect to t , we have

$$\frac{dx'}{dt} = -\omega x \sin \omega t + \omega y \cos \omega t + \frac{dx}{dt} \cos \omega t + \frac{dy}{dt} \sin \omega t.$$

Or, since $-x \sin \omega t + y \cos \omega t = y'$ [relation (vi), above], we have

$$\frac{dx'}{dt} = \omega y' + \frac{dx}{dt} \cos \omega t + \frac{dy}{dt} \sin \omega t. \quad \dots(viii)$$

Similarly,

$$\frac{dy}{dt} = -\omega x \cos \omega t - \omega y \sin \omega t - \frac{dx}{dt} \sin \omega t + \frac{dy}{dt} \cos \omega t.$$

Or, since $x \cos \omega t + y \sin \omega t = x'$ [relation (v) above], we have

$$\frac{dy}{dt} = -\omega x' - \frac{dx}{dt} \sin \omega t + \frac{dy}{dt} \cos \omega t. \quad \dots(ix)$$

And, of course,

$$\frac{dz'}{dt} = \frac{dz}{dt}. \quad \dots(x)$$

Differentiating expressions (viii), (ix) and (x) once again, with respect to t , we have

$$\frac{d^2 x}{dt^2} = \omega \frac{dy'}{dt} - \omega \frac{dx}{dt} \sin \omega t + \omega \frac{dy}{dt} \cos \omega t.$$

Since from relation (ix) above,

$$-\frac{dx}{dt} \sin \omega t + \frac{dy}{dt} \cos \omega t = \left(\frac{dy'}{dt} + \omega x' \right), \text{ we have}$$

$$\frac{d^2 x'}{dt^2} = \omega \frac{dy'}{dt} + \omega \left(\frac{dy'}{dt} + \omega x' \right) = 2\omega \frac{dy'}{dt} + \omega^2 x', \quad \dots(xi)$$

Similarly,

$$\frac{d^2 y'}{dt^2} = -\omega \frac{dx'}{dt} - \omega \frac{dx}{dt} \cos \omega t - \omega \frac{dy}{dt} \sin \omega t.$$

Since from relation (viii) above, $\frac{dx}{dt} \cos \omega t + \frac{dy}{dt} \sin \omega t = \left(\frac{dx'}{dt} - \omega y' \right)$,

$$\text{we have } \frac{d^2 y'}{dt^2} = -\omega \frac{dx'}{dt} - \omega \left(\frac{dx'}{dt} - \omega y' \right) = -2\omega \frac{dx'}{dt} + \omega^2 y'. \quad \dots(xii)$$

And

$$\frac{d^2 z'}{dt^2} = \frac{d^2 z}{dt^2}. \quad \dots(xiii)$$

It is thus clear from relations (xi) and (xii) above, that even though no force is acting on particle P in frame S , a force *seems* to be acting on it in frame S' , producing an acceleration in it. *Frame S' is, therefore, a non-inertial frame of reference.* (See also § 2.14).

2.14 NON-INERTIAL FRAMES—FICTITIOUS FORCES

We have seen that (i) a frame of reference in accelerated translational motion with respect to an inertial frame (§ 2.10 and (ii) a reference frame in uniform rotation about an inertial frame (§ 2.13), are both *non-inertial* frames. Let us consider them a little more in detail.

(i) **The accelerated frame of reference.** It has been shown in § 2.10 how if a force $\vec{F} = m\vec{a}$ be acting on a particle in an inertial frame S , the force acting on it in a reference frame S' , moving with an acceleration $\vec{a}_0$ with respect to S , in $\vec{F}' = (m\vec{a} - m\vec{a}_0)$. Frame S' is, therefore, a non-inertial frame since Newton's second law of motion, obviously, does not hold good in its case. For, putting $-m\vec{a}_0 = \vec{F}_0$, with $m\vec{a}$, of course, equal to $\vec{F}$, we have $-\vec{F}' = \vec{F} + \vec{F}_0$. And, as we have seen already, if $\vec{F}' = \vec{F}_0 = -m\vec{a}_0$, i.e., even when no force actually acts on the particle in the inertial frame S , a force equal to $\vec{F}_0$ appears to be acting on it in frame S' .

This force $\vec{F}_0 = m\vec{a}_0$ does not actually exist but appears to come into being purely in consequence of the acceleration of frame S' with respect to S . It is, therefore, a aptly called a **fictional** or a **pseudo (false) force**. And, clearly, it is equal to mass times the acceleration of the non-inertial frame, with its sign reversed.

Or, fictional force $\vec{F}_0 = \text{mass (acceleration of non-inertial frame, with sign reversed)}$.

$$\text{Or, } \vec{F}_0 = m(-\vec{a}_0) = -m\vec{a}_0$$

As will be easily seen, it is the force which, when added to the true force ($\vec{F}$) in the inertial frame S , gives the observed force $\vec{F}'$ in the non-inertial or accelerated frame S' .

This means, in other words, that Newton's second law of motion will also hold good in the non-inertial frame S' which will, therefore, also behave as an inertial frame of reference, if we add to the true force $\vec{F}$ a fictional force $\vec{F}_0 = -m\vec{a}_0$.

Examples of Fictional force.

A familiar example of a fictional force in an accelerated frame of reference is that on a particle at rest with respect to a lift which is descending with an acceleration equal to g , if m be the mass of the particle, the fictional force acting on it is $F_0 = -mg$. So that, the resultant force on it, as observed by an observer in the lift (the moving frame) is given by $F' = \text{true force acting on it, (i.e., } mg) + \text{the fictional force, } -mg$, Or, $F' = mg - mg = 0$, i.e., the particle is weightless and thus remains suspended in the air, —a condition that obtains in artificial satellites.

Exactly similarly, a body on the palm of our hand appears to our palm to be weightless during the small fraction of time that we are jumping down from a table on to the ground. For, here too, the body is initially at rest with respect to our palm which moves down with an acceleration equal to g .

Another interesting point about a fictional force is that it enables us to determine whether or not a given frame of reference is accelerated. For, if two frames be in uniform relative motion (zero acceleration), with respect to each other, they are obviously inertial frames, and it is impossible to find out which one is at rest and which one in motion. If, however, one of them be accelerated with respect to the other, fictional forces come into play inside the accelerated frame. These forces are obviously different from the true or real forces. For, unlike the latter, which go on diminishing rapidly with distance and are almost zero at a large distance from a body, the former arise precisely due to the accelerated motion of the frame of reference and go on increasing with its acceleration. So that, whereas the real forces on particles or bodies at large distances from other bodies are negligible, the fictional forces may have appreciable values. If, therefore, we find a body or

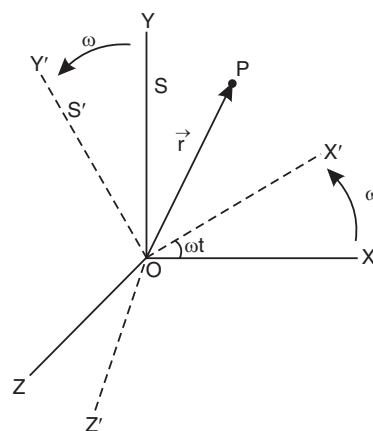


Fig. 2.11

a particle, far away from other material bodies, to be acted upon by an appreciable force, we can safely declare that its frame of reference is an accelerated one.

(ii) The rotating frame of reference-Coriolis acceleration and force. Let us now consider a reference frame S' (shown dotted) rotating with a uniform angular velocity $\vec{\omega}$ with respect to an inertial frame S (Fig. 2.11). To make matters simpler, let us assume the two frames of reference to have a common origin O and the axis of rotation of S' about S to be passing through O .

Suppose now we have a moving particle P (like, for instance, a fly or an ant walking on a rotating piece of cardboard). The value of its position vector $\vec{r}$ at any given instant t will, obviously, be the same in either frame of reference, though its components along the three axes will naturally be different.

If the particle (*i.e.*, the fly or the ant on the card board) be at rest with respect to the inertial frame S , it would appear to be moving with a relative linear velocity $-\vec{\omega} \times \vec{r}$ in the non-inertial, rotating frame S' . And, therefore, if its linear velocity in frame S be $\vec{v} = d\vec{r}/dt$, it will appear in frame S' to be $\vec{v}' = \frac{d\vec{r}}{dt} - \vec{\omega} \times \vec{r}$

$$\text{So that, we have } \left(\frac{d\vec{r}}{dt} \right)_{S'} = \left(\frac{d\vec{r}}{dt} \right)_S - \vec{\omega} \times \vec{r}. \text{ Or, } \vec{v}' = \vec{v} - \vec{\omega} \times \vec{r},$$

$$\text{whence, } \left(\frac{d\vec{r}}{dt} \right)_S = \left(\frac{d\vec{r}}{dt} \right)_{S'} + \vec{\omega} \times \vec{r}. \quad \dots(i)$$

$$\text{Or, } \vec{v} = \vec{v}' + \vec{\omega} \times \vec{r}.$$

This gives the linear velocity ($\vec{v}$) of the particle in the inertial frame S at any given instant in terms of its velocity ($\vec{v}'$) in the rotating frame S' and its position vector $\vec{r}$ at that instant.

Now, equation (i) above, relating the rate of change of a vector ($\vec{r}$ in this case) in two given frames of reference (S and S') is clearly a general one, applicable to any vector and may, therefore, be put in the form of an operator equation $\left(\frac{d}{dt} \right)_S = \left(\frac{d}{dt} \right)_{S'} + \vec{\omega} \times \vec{r}$ (ii)

Applying this to the velocity vector $\vec{v}$, in order to deduce a relationship between the accelerations of the particle in the two frames of reference, we have

$$\left(\frac{d\vec{v}}{dt} \right)_S = \left(\frac{d\vec{v}}{dt} \right)_{S'} + \vec{\omega} \times \vec{r}, \text{ i.e., } \frac{d\vec{v}}{dt} = \frac{d\vec{v}'}{dt} + \vec{\omega} \times \vec{v}.$$

Or, substituting the value of $\vec{v}$, we have

$$\left(\frac{d\vec{v}}{dt} \right)_S = \frac{d}{dt}[(\vec{v}' + \vec{\omega} \times \vec{r})]_{S'} + \vec{\omega} \times (\vec{v}' + \vec{\omega} \times \vec{r})$$

$$\text{Or, } \frac{d\vec{v}}{dt} = \frac{d\vec{v}'}{dt} + \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \left(\frac{d\vec{r}}{dt} \right)_S + \vec{\omega} \times \vec{v}' + \vec{\omega} \times (\vec{\omega} \times \vec{r}).$$

Clearly $d\vec{v}/dt$ is the acceleration of the particle, as observed in the inertial frame S . Denoting it by $\vec{a}$, and denoting the acceleration $d\vec{v}'/dt$ in the rotating frame S' by $\vec{a}'$, we have

$$\vec{a} = \vec{a}' + 2\vec{\omega} \times \vec{v} + \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad [\because (d\vec{r}/dt)_{S'} = \vec{v}']$$

Since the angular velocity $\vec{\omega}$ is uniform, $d\vec{\omega}/dt = 0$. We, therefore, have

$$\vec{a} = \vec{a}' + 2\vec{\omega} \times \vec{v}' + \vec{\omega} \times (\vec{\omega} \times \vec{r}).$$

Here, $2\vec{\omega} \times \vec{v}'$ is what is called Coriolis acceleration, after G. Coriolis, who first discovered it, and $\vec{\omega} \times (\vec{\omega} \times \vec{r})$, the familiar **centripetal acceleration**. So that, we have

$$\text{acceleration in the inertial frame} = \text{observed acceleration in the rotating frame} + \text{Coriolis acceleration} + \text{centripetal acceleration}.$$

Coriolis force

Hence, if m be the mass of the particle, we have force acting on the particle in the rotating frame $= m\vec{a}' = \vec{F}'$ and the true force acting on it in the inertial frame $= m\vec{a} = \vec{F}$.

$$\text{And, therefore, } m\vec{a}' = m\vec{a} - 2m\vec{\omega} \times \vec{v}' - m\vec{\omega} \times (\vec{\omega} \times \vec{r}).$$

$$\text{Or, } \vec{F}' = \vec{F} - 2m\vec{\omega} \times \vec{v}' - m\vec{\omega} \times (\vec{\omega} \times \vec{r}).$$

But, as we know, the force acting in a non-inertial frame, i.e.,

$$\vec{F}' = \vec{F} + \vec{F}_0$$

where $\vec{F}_0$ is the fictitious force acting in the non-inertial frame.

$$\text{So, that, here, fictitious force } \vec{F}_0 = -2m\vec{\omega} \times \vec{v}' - m\vec{\omega} \times (\vec{\omega} \times \vec{r}),$$

where $-2m\vec{\omega} \times \vec{v}'$ is the Coriolis force and $-m\vec{\omega} \times (\vec{\omega} \times \vec{r})$, the centrifugal force.

Thus, here, fictitious force $\vec{F}_0 = \text{Coriolis force} + \text{centrifugal force}$.

It will thus be seen that the centrifugal force is a fictitious force which acts on a particle at rest relative to a rotating frame of reference. Being responsible for keeping the particle at rest in the rotating frame, it is numerically equal to the centripetal force, $m\omega^2 \times \vec{r}_n$, where $\vec{r}_n$ is the distance of the particle from the centre of rotation (or the component of $\vec{r}$ normal to the axis of rotation) but is oppositely directed, i.e., outwards, away from the axis of rotation, Fig. 2.12).

It will also be seen that Coriolis force too is a fictitious force, which acts on a particle in motion relative to a rotating frame of reference. It is proportional to the angular velocity ($\vec{\omega}$) of the rotating frame and to the velocity $\vec{v}'$ of the particle relative to it. Its direction is always perpendicular to that of $\vec{v}'$ and is obtained by a rotation through 90° in the opposite sense to that of $\vec{\omega}$, as will be clear from Fig. 2.11.

It will obviously be zero if either $\vec{v}'$ or $\vec{\omega}$ is zero, i.e., if the particle is at rest relative to the rotating frame (in which case the only fictitious force acting on the particle will be the centrifugal force) or if the reference frame be a non-rotating one.

It will further be seen that particle P (Fig. 2.11) will move relative to the rotating frame S' in accordance with Newton's law of motion if we add to the true force $\vec{F}$, these two fictitious forces.

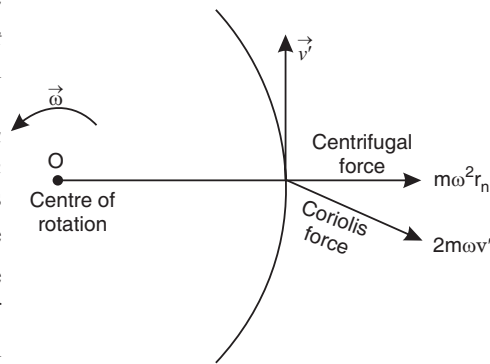


Fig. 2.12

2.15 EFFECTS OF CENTRIFUGAL AND CORIOLIS FORCES DUE TO EARTH'S ROTATION

Since the earth is rotating uniformly about its own axis, from west to east, a reference frame fixed on it is clearly a rotating frame of reference. As a result, the two fictitious forces, viz., the

Centrifugal force. and the Coriolis force must act on a particle, *at rest and in motion respectively*, relative to the rotating frame or reference. Let us see the effect of these two forces.

(i) **Effect of centrifugal force.** Let us consider a particle P at rest on the surface of the earth, in latitude ϕ (Fig. 2.13). Then, because the particle is at rest in the rotating frame of reference of the earth, *there is no Coriolis force acting on it*. The only fictitious force effective is therefore, the centrifugal force $= m\omega \times (\omega \times \mathbf{R}_n)$, where $\mathbf{R}_n$ is the component of the radius R of the earth, perpendicular to its axis (or the distance of the particle P from the axis). Hence, if the *observed or apparent acceleration of the particle*, in latitude ϕ , be directed towards C , and its *true acceleration*, g , towards the centre (O) of the earth, we have

$$g_\phi = g - \omega' \times (\omega \times R_n).$$

Now, if we take the axes, OY and OX , along and perpendicular to ω respectively, with $\hat{i}$ and $\hat{j}$ as the unit vectors along OX and OY as usual, we have

$$\vec{g} = -g(\cos\phi\hat{i} + \sin\phi\hat{j}), \vec{\omega} = \omega\hat{j} \text{ and } \vec{R}_n = R\cos\phi\hat{i}.$$

Substituting in the expression for g_ϕ above, therefore, we have

$$\begin{aligned} \vec{g}_\phi &= -g(\cos\phi\hat{i} + \sin\phi\hat{j}) - \omega\hat{j} \times (\omega\hat{j} \times R\cos\phi\hat{i}) \\ &= -g(\cos\phi\hat{i} + \sin\phi\hat{j}) + \omega^2 R\cos\phi\hat{i}, \end{aligned}$$

i.e., the magnitude of the apparent acceleration is

$$g_\phi = \sqrt{(g\cos\phi - \omega^2 R\cos\phi)^2 + g^2\sin^2\phi}.$$

Since ω is small, we may neglect terms involving ω^4 .

So that, we have

$$g_\phi = \sqrt{(g^2\cos^2\phi - 2g\omega^2 R\cos^2\phi + g^2\sin^2\phi)} = \sqrt{g^2 - 2g\omega^2 R\cos^2\phi}.$$

Or,

$$g_\phi = g \left(1 - \frac{2\omega^2 R\cos^2\phi}{g} \right)^{\frac{1}{2}} = g \left(1 - \frac{\omega^2 R\cos^2\phi}{g} \right)$$

very nearly,

$$i.e., \quad g_\phi = g - \omega^2 R\cos^2\phi.$$

This acceleration, as mentioned above, is directed towards C instead of O , the centre of the earth. So that, if the angle that this *apparent direction* PC makes with the *true direction* PO be θ , we have

$$\theta = \tan^{-1} \frac{g\cos\phi - \omega^2 R\cos\phi}{g\sin\phi} = \tan^{-1} \left[\left(1 - \frac{\omega^2 R}{g} \right) \cot\phi \right].$$

Thus, the effect of centrifugal force, due to rotation of the earth, is to reduce the effective value of g on its surface as also to slightly change its direction from the truly vertical,—towards the north and the south respectively in the two hemispheres.

Obviously, θ gives the deflection of a plumb line from the true direction of the gravitational force and will have its maximum value in latitude $\phi = 45^\circ$.

(ii) **Effect of Coriolis force.** In case a body is in motion relative to the rotating frame of reference of the earth, the fictitious Coriolis force comes into play and two cases of interest arise :

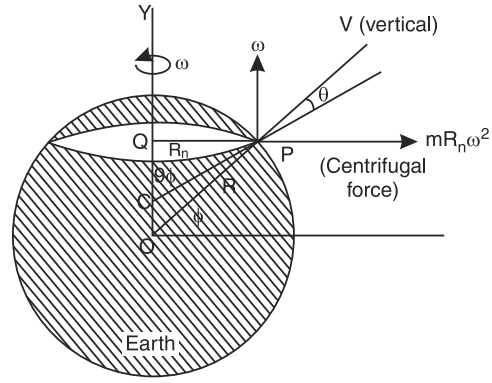


Fig. 2.13

(a) when the body is just dropped from rest so as to fall freely under the action of the gravitational force and (b) when it is given a large horizontal velocity (case of a projectile).

In case (a), the horizontal component of the Coriolis force acting on the freely falling body deflects it a little from its truly vertical path. The vertical component, obviously, produces no south deflection but only affects the value of g .

To calculate this deflection or displacement of the falling body, let us take the axes of x , y and z along the *east*, *north* and *vertically upwards* respectively and $\hat{i}$, $\hat{j}$ and $\hat{k}$ as the unit vectors along these axes. Then, if $\vec{v}$ be the velocity acquired by the body in time t taken by it to fall through a height h , we have

$$\vec{v} = -v\hat{k} \quad (\because \text{the velocity is directed downwards})$$

and $\omega = \omega \cos \phi \hat{j} + \omega \sin \phi \hat{k}$, where ϕ is the *latitude* at the place.

$$\begin{aligned} \therefore \text{Coriolis acceleration, say, } \vec{a}_c &= -2\vec{\omega} \times \vec{v} \\ &= -2\omega (\cos \phi \hat{j} + \sin \phi \hat{k}) \times (-v\hat{k}) \\ &= 2\omega v (\cos \phi \hat{i} - 0) = 2\omega v \cos \phi \hat{i}, \end{aligned}$$

i.e., the Coriolis acceleration on the body in latitude ϕ is $2\omega v \cos \phi$ along the positive direction of the axis of x or is directed towards the east.

The equation of motion of the body may, therefore, be put as

$$d^2x/dt^2 = 2\omega v \cos \phi.$$

Since v is the velocity acquired by the body in time t , we have $v = 0 + gt = gt$, since its initial velocity is zero and the acceleration due to gravity, g . So that,

$$d^2x/dt^2 = 2\omega gt \cos \phi.$$

$\therefore$ x -component of the velocity of the body, say,

$$v_x = dx/dt = \int 2\omega gt \cos \phi dt.$$

Or, $v_x = 2\omega g \cos \phi t^2/2 + C$, where C is the constant of integration.

Since v_x is zero at the very start, when $t = 0$, we have $C = 0$.

And $\therefore$ $v_x = \omega g \cos \phi t^2$.

Integrating once again, we have

$$\begin{aligned} \text{displacement along the axis of } x, \text{ i.e., } x &= \int \omega g \cos \phi t^2 dt \\ &= \omega g \cos \phi \left(\frac{t^3}{3} \right) + C'. \end{aligned}$$

where C' is another **constant of integration**.

Again, since at $t = 0$, there is no displacement, *i.e.*, $x = 0$, we have $C' = 0$.

Hence, $x = \frac{1}{3} \omega g \cos \phi t^3$.

Now, t is the time taken by the body to fall through a height h . So that, its initial velocity being zero, we have $h = \frac{1}{2}gt^2$ whence, $t = \sqrt{2h/g}$. Substituting this value of t in the expression for x above, we have

$$x = \frac{1}{3} \omega g \cos \phi (2h/g)^{\frac{3}{2}} = \left(\frac{8}{9g} \right)^{\frac{1}{2}} h^{\frac{3}{2}} \omega \cos \phi.$$

Thus, the horizontal displacement of the body due to Coriolis force (i) in latitude ϕ is equal to $(8/9g)^{1/2} h^{3/2} \omega \cos \phi$ and (ii) at the equator, since $\phi = 0$, or $\cos \phi = 1$, it is equal to $(8/9g)^{1/2} h^{3/2} \omega$

(i.e., the maximum). And it is always directed along the positive direction of the x-axis or towards the east.

In case (b), if the horizontal velocity of the body, (or the projectile) be sufficiently large, so that it covers fairly large horizontal distances, the small *Coriolis force* gets sufficient time to act upon it, making the position vector turn at a constant rate of $-\omega \sin \phi$. Since in the Northern hemisphere, ϕ is positive, this rotation, as viewed from above, is *clockwise* (and hence the projectile gets deflected towards the right) and in the Southern hemisphere, it is *anticlockwise* (and the projectile thus gets deflected towards the left). This is known as **Ferrel's law**.

Example 11. Calculate the fictitious force and the observed (or total) force on a body of mass 5 kg in a frame of reference moving (i) vertically upwards, (ii) vertically downwards, with an acceleration of 4 m/s^2 . (Take $g = 9.8\text{ m/s}^2$).

Solution. Taking the earth to be an inertial frame of reference and the upward direction as positive, we have

Case (I) True weight of the body or the true force acting on it, i.e., $F = mg = 5(-9.8) = -49.0$ newton, i.e., 49.0N in the downward direction.

And, the fictitious force acting on it, i.e., $F_0 = \text{mass (reversed acceleration of the non-inertial frame)} = m(-a_0) = 5(-4) = -20$ newton, i.e., 20N in the downward direction.

$\therefore$ observed or total force on the body, $\vec{F}' = \vec{F} + \vec{F}_0 = 49.0 + 20.0 = 69.0\text{ N}$ in the downward direction, (i.e., the same as in a lift or elevator moving upwards with an acceleration of 4 m/s^2)

Thus, the fictitious force in the body is 20 N and the observed (or total) force on it, 69 N, both acting downwards.

Case (II). True force acting on the body, as before, $\vec{F} = mg = 5(-9.8) = -49.0$ newton = 49.0N downwards.

And, the fictitious force acting on the body, $\vec{F}_0 = m(-a_0) = 5[-(-4)]$
 $= 20.0\text{ N}$ upwards.

$\therefore$ observed or total force on the body. $\vec{F}' = \vec{F} + \vec{F}_0 = -49.0 + 20.0$
 $= -29.0$ newton
 $= 29.0\text{ N}$ downwards,

(i.e., the same as in a lift or elevator, moving downwards with an acceleration of 4 m/s^2).

Thus, in this case, fictitious force on the body is 20N upwards and the observed (or total) force on it, 29 N downwards.

Example 12. Calculate the total force acting on a freely falling body of mass 5 kg with reference to a frame moving with a downward acceleration of 2 ms^{-2} .

Solution. The force $\vec{F}'$ acting on a body in a non-inertial frame is given by $\vec{F}' = \vec{F}_i - \vec{F}_0$ where $\vec{F}_i$ is the force on the same body in an inertial frame and $\vec{F}_0$ is the fictitious force due to the accelerated motion of the non-inertial frame. As the body is falling freely, downward force on it in the inertial frame of the earth $\vec{F}_i = 0$

$$\therefore \vec{F}' = -\vec{F}_0 \quad \text{or} \quad \vec{F}' = -m\vec{a}_0$$

where $\vec{a}_0$ is the acceleration of the non-inertial frame and m the mass of the body.

As the reference frame is moving downward with acceleration of 2 m/s^2 $\vec{a}_0 = -2\text{ ms}^{-2}$

$$\therefore \vec{F}' = -m\vec{a}_0 = -(-2 \times 5) = +10\text{ N}$$

The positive sign indicates that the fictitious force is acting upward.

Example 13. A frame of reference is moving with an acceleration of 5 m/sec^2 downward. Find the apparent force and total force acting on a body of mass 10 kg falling freely relative to the frame.

Solution. Apparent force = Total force = $-(-5 \times 10) = 50 \text{ N}$ (upward).

Example 14. Calculate the effective weight of an astronaut ordinarily weighing 60 kg when his rocket moves vertically upward with $5g$ acceleration.

Solution. As the rocket moves vertically upward with an acceleration $5g$, it is a non-inertial frame and therefore the total force on the astronaut is given by $\vec{F}' = \vec{F}_i - \vec{F}_0$ where $\vec{F}_i$ is the force on the astronaut in an inertial frame and $\vec{F}_0$ is the fictitious force on the astronaut due to the acceleration of the rocket.

$$\text{Now} \quad \vec{F}_i = 60 \text{ kg. wt} = 60.g \text{ N}$$

$$\text{and} \quad \vec{F}_0 = -m\vec{a}_0 = -60 \times 5.g \text{ N} = -300.g \text{ N}$$

$\therefore$ Effective weight of the astronaut

$$\begin{aligned} \vec{F} &= \vec{F}_i - \vec{F}_0 = 60g - (-300g) \\ &= 360g \text{ N} = 360 \text{ kg.} \end{aligned}$$

Example 15. How much faster than its present speed should the earth rotate so that bodies lying on the equator may fly off into space?

Solution. The centrifugal force acting on a body of mass m at the equator is equal to $mR\omega^2$ where R is the radius of the earth and ω its angular velocity. If g is the acceleration due to gravity when the earth is at rest, then g' the acceleration when the earth has an angular velocity ω is given by

$$g' = g - R\omega^2 \cos^2 \lambda = g - R\omega^2 \quad [\because \lambda = 0, \cos \lambda = 1 \text{ and } \cos^2 \lambda = 1]$$

For the body to fly off into space the new angular speed ω' should be such that

$$g - R\omega'^2 = 0 \quad \text{or} \quad \frac{R\omega'^2}{g} = 1 \quad \dots(i)$$

$$\text{but} \quad \frac{R\omega^2}{g} = \frac{1}{291} \quad \dots(ii)$$

$$\text{Dividing (i) by (ii), we have} \quad \frac{\omega'^2}{\omega^2} = 291 \quad \text{or} \quad \frac{\omega'}{\omega} = \sqrt{291} = 17.06$$

Hence the earth should have about seventeen times the present angular velocity so that bodies lying on the equator may fly into space.

Example 16. If the earth were to cease rotating about its axis what will be the change in the value of g at a place of latitude 45° assuming the earth to be a sphere of radius $6.38 \times 10^8 \text{ cm}$?

Solution. We know that $g' = g - R\omega^2 \cos^2 \lambda$ or $g - g' = R\omega^2 \cos^2 \lambda$

$$\text{Now} \quad \lambda = 45^\circ; R = 6.38 \times 10^8 \text{ cm}, \omega = 2\pi/60 \times 60 \times 24$$

$$\begin{aligned} \text{Substituting, we have} \quad g - g' &= \frac{6.38 \times 10^8 \times 4\pi^2}{60 \times 60 \times 60 \times 24 \times 24} \cos^2 45^\circ \\ &= 3.372 \times \frac{1}{2} = 1.686 \text{ cm/sec}^2. \end{aligned}$$

Example 17. Find the magnitude and direction of Coriolis force that acts on a 800 kg van running due north at 144 km hr^{-1} at a place where latitude is 30°N . (G.N.D.U. 2004)

Solution. Mass of the van $m = 800 \text{ kg}$

Velocity $\vec{v} = 144 \text{ km hr}^{-1} = \frac{144 \times 1000}{60 \times 60} = 40 \text{ ms}^{-1} \text{ due North}$

Latitude of the place $\lambda = 30^\circ \text{ N} \quad \therefore \sin \lambda = \sin 30 = \frac{1}{2}$

$$\omega \text{ for the earth} = \frac{2\pi}{24 \times 60 \times 60} \text{ rad/sec.}$$

The X -axis is taken towards the East, Y -axis towards the North and Z -axis in the vertically upward direction.

Coriolis force is given by $\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}_a)$

Horizontal component of Coriolis force

$$(\vec{F}_{\text{cor}})_{\text{Horizontal}} = +2m\omega \sin \lambda (v_y \hat{i} - v_x \hat{j})$$

and vertical component

$$(\vec{F}_{\text{cor}})_{\text{Vertical}} = 2m\omega v_x \cos \lambda \hat{k}$$

As the van is running due North (along $+Y$ -axis) the component of its velocity due East (along $+X$ -axis) *i.e.*, $v_x = 0$ and $v_y = v_a = \vec{v}$

$\therefore$ Vertical component of Coriolis force $= 0$

Horizontal component of Coriolis force,

$$\begin{aligned} (\vec{F}_{\text{cor}})_H &= +2m\omega \sin \lambda v_y \hat{i} \\ &= 2 \times 800 \times \frac{2\pi}{24 \times 60 \times 60} \times \frac{1}{2} \times 40 \\ &= 7 \text{ Newton} \end{aligned}$$

This force will act along $+\hat{i}$, *i.e.*, towards East.

Note. If the latitude of the place is 30°S , the force will act along $-\hat{i}$, *i.e.*, towards the West.

Example 18. A bullet is fired horizontally with a velocity 500 ms^{-1} at 60°N colatitude, (a) Eastwards (b) Northwards. Calculate the magnitude of the Coriolis acceleration due to rotation of earth. (P.U. 2001)

Solution. Velocity of the bullet $\vec{v} = 500 \text{ ms}^{-1}$

Co-latitude $\phi = 60^\circ \quad \therefore \text{Latitude } \lambda = (90 - \phi) = 30^\circ \text{ and } \sin \lambda = \sin 30 = \frac{1}{2}.$

The X -axis is taken towards the East, Y -axis towards the North and Z -axis in the vertically upward direction.

Coriolis acceleration is given by $\vec{a}_a = -2\vec{\omega} \times \vec{v}_a$

Horizontal component of Coriolis acceleration $(\vec{a}_a)_H = 2\omega \sin \lambda (v_y \hat{i} - v_x \hat{j})$

and Vertical component of Coriolis acceleration $(\vec{a}_a)_V = 2\omega v_x \cos \lambda \hat{k}$

(a) Bullet fired Eastward. Then $v_x = \vec{v} = 500 \text{ ms}^{-1}$; $v_y = 0$

$\therefore$ Horizontal component of Coriolis acceleration.

$$\begin{aligned} (\vec{a}_a)_H &= -2\omega \sin \lambda v \hat{j} \\ &= 2 \times \frac{2\pi}{24 \times 60 \times 60} \times \frac{1}{2} \times 500 = 0.036 \text{ ms}^{-1} \end{aligned}$$

along $-\hat{j}$ *i.e.*, $-Y$ direction or towards South.

Vertical component of Coriolis acceleration

$$\begin{aligned}
 (\vec{a}_a)_V &= 2\omega v_x \cos \lambda \hat{k} \\
 &= 2 \times \frac{2\pi}{24 \times 60 \times 60} \times \frac{\sqrt{3}}{2} \times 500 = 0.036\sqrt{3}
 \end{aligned}$$

along $+\hat{k}$ i.e., $+Z$ direction or vertically upwards.

$\therefore$ Magnitude of the resultant acceleration

$$|\vec{a}_a| = |(\vec{a}_a)_H|^2 + |(\vec{a}_a)_V|^2 = \sqrt{(0.036)^2 + (0.036\sqrt{3})^2} = 0.072 \text{ ms}^{-1}$$

(b) Bullet fired Northwards. Then $v_y = \vec{v} = 500 \text{ ms}^{-1}$; $v_x = 0$

$\therefore$ Horizontal component of coriolis acceleration $(\vec{a}_a)_H = 2\omega \sin \lambda v \hat{i} = 0.036 \text{ ms}^{-1}$ along $\hat{i}$ i.e., $+X$ direction.

Vertical component of coriolis acceleration $(\vec{a}_a)_V = 2\omega v_x \cos \lambda \hat{k} = 0$

$\therefore$ Resultant acceleration $= 0.036 \text{ ms}^{-1}$ along $+X$ direction.

Example 19. Calculate the magnitude and direction of Coriolis acceleration of a rocket moving with a velocity of 2 km s^{-1} at 60° South latitude.

Solution. For a body moving in a vertical direction, Coriolis force is given by

$$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}) = -2m \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & \omega_y & \omega_z \\ 0 & 0 & \dot{z} \end{vmatrix}$$

$$\therefore \vec{F}_{\text{cor}} = -2m\omega_y \dot{z} \hat{i}$$

For a rocket moving vertically *upwards* at 60° South latitude (λ)

$$\dot{z} = +v \quad \text{and} \quad \omega_y = -\omega \cos \lambda \quad \therefore \vec{F}_{\text{cor}} = +2m\omega v \cos \lambda \hat{i}$$

The Coriolis force, therefore acts towards the positive direction of x -axis i.e., towards the East. The magnitude of Coriolis acceleration $= \omega v \cos \lambda$

$$= 2 \times \frac{2\pi}{24 \times 60 \times 60} \times 2 \times 10^3 \times \cos 60 = 14.58 \times 10^{-2} \text{ m/s}^2.$$

Example 20. Find latitude of a place where the plane of vibration of Foucault's pendulum rotates once a day.

Solution. The period of rotation of a Foucault's pendulum at a latitude λ is given by

$$T = \frac{2\pi}{\omega \sin \lambda}. \quad \text{Now } \frac{2\pi}{\omega} = 24 \text{ hours}$$

For a place where plane of vibration of a Foucault's pendulum rotates once a day $T = 24$ hours.

$$\therefore \sin \lambda = 1 \text{ or } \lambda = \frac{\pi}{2}, \text{ i.e., at a latitude } \lambda = \frac{\pi}{2} \text{ or at the poles.}$$

Example 21. Calculate the rate of rotation of the plane of oscillation of a pendulum at latitude 30° and hence obtain the time it will take to turn through (i) a full right angle and (ii) 60° .

(P.U., 2001)

Solution. At a latitude $\lambda = 30^\circ$; $\sin \lambda = \frac{1}{2}$

$$\therefore \text{The period of rotation } T = \frac{2\pi}{\omega \sin \lambda} = \frac{2\pi}{\omega \frac{1}{2}} = 24 \times 2 = 48 \text{ hours.}$$

$$\text{Hence rate of rotation} = \frac{2\pi}{48} \text{ radian/hour}$$

and time taken to turn through (i) a full right angle or $\pi/2$ radian = $\frac{\pi/2}{2\pi/48} = \frac{48}{4} = 12$ hours

and (ii) through $60^\circ = \frac{\pi/3}{2\pi/48} = \frac{48}{6} = 8$ hours.

Example 22. Imagine two non-inertial frames S' and S'' to be initially coincident with an inertial frame S , with S' at rest and S'' having a velocity v along the axis of x and both subjected to the same acceleration a along this axis (Fig. 2.14). Obtain

- the positions of the origins O' and O'' of the two non-inertial frames with respect to the origin O of the inertial frame after time t ;
- the relations between the respective positions x' and x'' of a particle, in frames S' and S'' , and its position x in frame S ;
- the equation of motion of a particle of mass m , subjected to a constant force F in frame S and hence also its equations of motion in frames S' and S'' , indicating any point of interest the emerges ;
- the relative velocity of frames S'' with respect to frame S' .

What conclusion, if any, do you draw from the results obtained in regard to non-inertial frames, in general, moving with a constant relative velocity ?

Solution. (i) Initially, the origins O , O' and O'' of the three reference frames are coincident. Therefore, the initial velocity of frame S' being zero and its acceleration along the x -axis, a , the position vector of its origin O' after time t is given by

$$\vec{OO'} = \vec{r'_0} = \frac{1}{2}at^2\hat{i},$$

where $\hat{i}$ is the unit vector along the axis of x .

i.e., displacement of O' from O , along the x -axis, is $x' = \frac{1}{2}at^2$,

(Fig. 2.14).

And, since frame S'' has an initial velocity $\vec{v}$ along the x -axis, as also an acceleration $\vec{a}$ along it, its position after time t is given by $\vec{OO'} = \vec{r'_0} = \left(vt + \frac{1}{2}at^2\right)\hat{i}$

i.e., displacement of O'' from O along the x -axis is $x'' = vt + \frac{1}{2}at^2$.

(ii) If the positions of a particle in frames S , S' and S'' be respectively x , x' and x'' , we clearly have $x' = x - x'_0 = x - \frac{1}{2}at^2$ and $x'' = x - x''_0 = x - vt - \frac{1}{2}at^2$.

(iii) The equation of motion of a particle of mass m , subjected to a constant force $\vec{F}$, in frame S , is, obviously, $\vec{F} = m\frac{d^2\vec{r}}{dt^2}$, where $\vec{r}$ is the position vector of the particle in frame S .

Similarly, if $\vec{r}'$ and $\vec{r}''$ be the position vectors of the particle in frames

S' and S'' respectively, the equation of motion of the particle in frame S' is

$$\vec{F'} = m\frac{d^2\vec{r'}}{dt^2}.$$

and its equation of motion in frames S'' , $\vec{F''} = m\frac{d^2\vec{r''}}{dt^2}$.

In terms of components along the three Cartesian coordinate axes, we have in frame S'

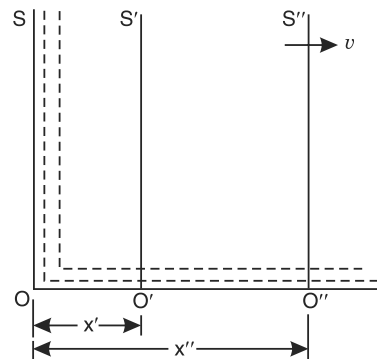


Fig. 2.14.

$$F'_x = m \frac{d^2 x'}{dt^2}, \quad F'_y = m \frac{d^2 y'}{dt^2} \quad \text{and} \quad F'_z = m \frac{d^2 z'}{dt^2}.$$

Since there is motion only along the x -axis (with $y' = y'' = y$ and $z' = z'' = z$), we need consider only the x -component. So that, we have

$$F'_{x0} = m \frac{d^2 x'}{dt^2} = m \frac{d^2}{dt^2} \left(x - \frac{1}{2} at^2 \right) = m \frac{d^2 x}{dt^2} = ma.$$

Clearly, $m \frac{d^2 x}{dt^2}$ is the x -component of force F on the particle in frame S . Denoting it by F_x , we, therefore, have

$$F'_x = F_x - ma \quad \dots(i)$$

Proceeding exactly similarly, we obtain

$$\text{in frame } S'', \quad F''_x = m \frac{d^2 x''}{dt^2} = m \frac{d^2}{dt^2} \left(x - u - \frac{1}{2} at^2 \right) = m \frac{d^2 x}{dt^2} - ma. \quad [\because v = \text{constant}]$$

$$\text{Or,} \quad F''_x = m \frac{d^2 x}{dt^2} - ma = F_x - ma. \quad \dots(ii)$$

It will thus be seen that the same fictitious force ma appears in both the transformation equations I and II, in view, no doubt, of the *same acceleration (a) of the two frames S' and S'' with respect to S .*

As will be readily seen from the transformation equation (i) and (ii) for frames S' and S'' respectively under (iii) above, *the applied and fictitious forces in frame S' are respectively equal to those in frame S'' .*

(iv) The relative velocity of frame S'' with respect to frame S' is obviously given by

$$\vec{v}''_2 = \frac{d\vec{r}'}{dt} - \frac{d\vec{r}''}{dt} = (v + at)\hat{i} - (at)\hat{i} = v\hat{i}.$$

What is true of the respective equality of the applied and fictitious forces in frames S' and S'' [as seen under (iii) above] may be said to be true, in general, in the case of all non-inertial frames moving with a constant relative velocity. This follows from the fact that all such frames, although non-inertial with respect to S , *are inertial with respect to each other in view of their constant relative velocity and, therefore, zero relative acceleration. The applied and fictitious forces in them are thus governed by Galilean invariance.*

Example 23. A mass of 1 kg is hurled horizontally due north with a velocity of 0.5 km per second in latitude 30° N. Obtain the magnitude and direction of the Coriolis force acting on the mass and its deflection in consequence thereof in a time-interval of half a second.

Solution. Taking the axes of x , y and z along the east, the north and vertically upwards respectively, with $\hat{i}$, $\hat{j}$ and $\hat{k}$ as the respective unit vectors along the three axes, as usual, we have
velocity of mass, $\vec{v} = 500 \hat{j}$ m/s.

Now, the *angular velocity vector of the earth*, $\vec{\omega}$, due to the spinning around of the earth from west to east, is directed along the axis. It, therefore, lies towards, the north at 30° with the horizontal in the given altitude $\phi(30^\circ \text{ N})$. We therefore have

$$\vec{\omega} = \omega \cos \phi \hat{j} + \omega \sin \phi \hat{k} = \omega (\cos 30^\circ \hat{j} + \sin 30^\circ \hat{k}).$$

where ω , as we know, is equal to $2\pi/24 \times 60 \times 60 = 0.727 \times 10^{-5}$ radian/s.

$$\begin{aligned} \therefore \text{Coriolis acceleration of the mass} &= -2\vec{\omega} \times \vec{v} = -2\omega (\cos 30^\circ \hat{j} + \sin 30^\circ \hat{k}) \times 500 \hat{j} \\ &= 2(0.727 \times 10^{-5}) \times \frac{1}{2} \times 500 \hat{i} = 36.35 \times 10^{-5} \text{ m/s}^2. \end{aligned}$$

(Or, we could have obtained the Coriolis acceleration directly from the relation $a_c = 2\omega v \sin \phi$, since the *effective angular velocity* of the earth, here, is $\omega \cos (90 - \phi) = \omega \sin \phi$).

Because the mass is 1 kg, we have Coriolis force acting on the mass

$$= 1 \times 36.35 \times 10^{-3} = 36.35 \times 10^{-3} \text{ N towards the east.}$$

$$\therefore \text{Deflection of the mass due to this force in } \frac{1}{2}s = \frac{1}{2} \times 36.35 \times 10^{-3} \times \left(\frac{1}{2}\right)^2 = 4.544 \times 10^{-3} \text{ m.}$$

2.16 FOUCAULT'S PENDULUM—DEMONSTRATION OF EARTH'S ROTATIONS

A Foucault's pendulum is just a simple pendulum, with a massive *bob* carried by a very long wire suspended from a rigid support, so as to have *equal freedom of oscillation in any direction*.

It constitutes a simple device to demonstrate the rotation of the earth about its own axis and was first used by Foucault for a public demonstration of the same in the year 1851, in Paris. The suspended mass in his case was 28 kg and the suspension wire about 70 metre long, so that the time-period of the pendulum was nearly 17 sec. Such a pendulum, once set oscillating, continues to oscillate for a long enough time.

The theory underlying the experiment is as follows :

If the pendulum (Fig. 2.15) were to be set up at the North pole of the earth, it may oscillate as a simple pendulum in a vertical plane which remains fixed in an inertial or Galilean frame of reference. Since the earth under it rotates from west to east with an angular velocity ω , the plane of oscillation of the pendulum would appear to an observer on the surface of the earth to be turning with an angular velocity $-\omega$, i.e., in the opposite direction (from east to west) to that of the earth.

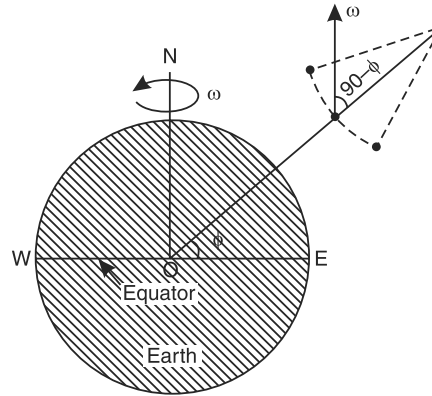


Fig. 2.15

It is by no means necessary, however, that the pendulum should be situated right at one of the poles. An apparent rotation of its plane of oscillation, due to the earth's rotation may equally well be observed in *any latitude*, except, of course, at the equator. For, it can be shown that, *if the oscillations be small, the bob describes an ellipse with its centre at the origin or the position of rest of the pendulum. The rotation of the earth (a rotating frame of reference) merely results in a Coriolis force coming into play due to the horizontal motion of the bob relative to it.* Acting, as usual, in a direction perpendicular to that of the velocity of the bob, it causes precession of the plane of its oscillation, which thus rotates with an angular velocity $-\omega \sin \phi$ in latitude ϕ .

Actually, however, there are two superimposed rotations of the elliptical orbit of the pendulum, one depending upon the area of the elliptical orbit and hence called the *area effect* and the other, on the angular velocity, $-\omega \sin \phi$, due to rotation of the earth and called the *Foucault effect*. The former is a rotation in the *same* sense in which the ellipse is described (and will get reversed in sign if that sense is reversed) and the latter, a rotation in a *definite direction*, viz., *clockwise in the Northern, and anticlockwise in the Southern, hemisphere*.

Since we are interested only in the rotation of the ellipse in consequence of the earth's rotation or its Foucault rotation, we must eliminate the *area effect* which, in the absence of suitable precautions, may be so much greater than the *Foucault effect* as to completely mask it. This is usually done by setting the pendulum in motion by *first drawing it a little to one side with the help of a piece of thread tied to it and then burning the thread*. The angular velocity of the ellipse (i.e., its rate of precession) due to the area effect is then negligible compared with its Foucault angular velocity, $-\omega \sin \phi$. So that, in the case of a pendulum, set oscillating in this manner, the precession or the rotation of its orbit may be taken to be due entirely to Foucault effect i.e., due to rotation of the earth alone.

Obviously enought, the time taken by one full rotation of the plane of oscillation of the pendulum is given by $T = 2\pi/(\text{Foucault angular velocity}) = 2\pi/\omega \sin \pi$. Or, since time taken by the earth to complete one rotation about its axis is 24 hours, we have $2\pi/\omega = 24$ hours and, therefore, $T = 24 \sin \phi$ hours.

As already mentioned, the rotation of the plane of oscillation of the pendulum will be clockwise in the Northern hemisphere and anticlockwise in the Southern hemisphere. There is thus clearly a *relative motion between the plane of oscillation of the pendulum bob and the earth is either hemisphere*. This relative angular shift of the plane of oscillation is measured over a period of days to demonstrate the rotation of the earth about its axis.

Among other methods of demonstrating the rotation of the earth may be mentioned (i) the observation of the *fixed stars* which, in fact, gave us the very first indication regarding the earth's rotation about its axis, (ii) the observation of any fictitious forces, indicating the presence of an accelerated or a rotating frame of reference, as mentioned already under 2.14; and (iii) the familiar gyroscope.

2.17 TIME-PERIOD AND ORBITAL SPEED OF A SATELLITE

A satellite is, in general, a smaller body revolving around a much larger body. Until recently, we had only *natural satellites*, like the earth revolving around the sun and the moon revolving around the earth in their respective orbits. The earth is thus a satellite of the sun and the moon, a satellite of the earth. We now have also *man-made* or *artificial* satellites, placed in orbit around the earth, with the help of powerful rockets. The principle underlying is the same in either case, *viz.*, that the gravitational attraction on the satellite due to the larger body (or the primary) pulling it *inwards*, supplies the centripetal force and this is just balanced by the (fictitious) centrifugal force on it, pulling it *outwards*, as shown in Fig. 2.16.

Thus, if m be the *mass of the satellite*, M , the *mass of the earth* r , the distance between the earth and the satellite (*i.e.*, between their centres), we have

gravitational pull of the earth on the satellite (i.e., the centripetal force), inwards = $\frac{mM}{r^2} G$, where G is the *gravitational constant*.

And *centrifugal force on the satellite*, outwards = mv^2/r , where v is uniform speed of the satellite around the earth.

It ω be the *angular velocity* of the satellite, so that $v = r\omega$, we have

$$\frac{mMG}{r^2} = \frac{mv^2}{r} = \frac{m(r\omega)^2}{r} = mr\omega^2$$

whence,

$$\omega^2 = MG/r^3. \quad \text{Or,} \quad \omega = \sqrt{MG/r^3}.$$

Since $r = (R + h)$, where R is the *radius of the earth* and h , the distance of the satellite from the surface of the earth, we have

$$\omega = \sqrt{MG/(R + h)^3}.$$

Now, if g be the acceleration due to gravity on the surface of the earth, we have

$$g = MG/R^2 \quad \text{or} \quad MG = gR^2.$$

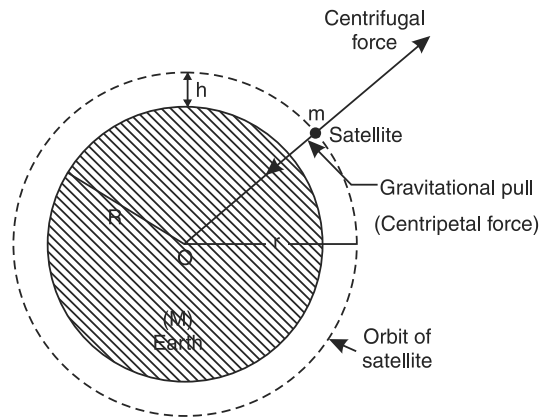


Fig. 2.16

Substituting this value of MG in the expression for ω above, we have

$$\omega = \sqrt{gR^2/(R+h)^3}.$$

And, therefore, *time-period of the satellite*,

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}}.$$

In case the orbit of the satellite lies close to the earth, *i.e.*, if $h \ll R$, we have $r = (R+h) = R$. So that,

$$T = 2\pi \sqrt{\frac{R^3}{gR^2}} = 2\pi \sqrt{\frac{R}{g}}$$

Let us calculate the orbital speed of such a satellite.

Remembering that, in this case, $r = (R+h) = R$ (the satellite being close to the earth), we have

$$r = r\omega = R\omega = R\sqrt{gR^2/R^3} = R\sqrt{g/R} = \sqrt{gR^2/R} = \sqrt{gR}.$$

So, that, taking $R = 6.4 \times 10^8$ cm and $g = 980$ cm/sec, we have

$$r = \sqrt{980 \times 6.4 \times 10^8} = 7.92 \times 10^5 \text{ cm/s}^2 = 7.92 \text{ km/s}.$$

Thus, *the orbital speed of an earth satellite revolving close to the earth's surface* is nearly 8 km/s.

Example 24. Find the latitude at which the plane of vibration of Foucault's pendulum does not rotate at all. (Pbi. U., 2003; P.U. 2003)

Solution. When the plane of vibration of Foucault's pendulum does not rotate at all its time period is infinite.

$\therefore$ The period of rotation $T = \infty$

and $\frac{2\pi}{\omega \sin \lambda} = \infty$ or $\lambda = 0$

Thus the plane of vibration of a Foucault's pendulum does not rotate at all at the Equator.

Example 25. Foucault's pendulum is oscillating along NS direction at a place where latitude is 30° . What time must elapse before the pendulum starts oscillating along NE-SW direction ?

Solution. At a latitude

$$\lambda = 30^\circ, \sin \lambda = \sin 30 = \frac{1}{2}$$

$\therefore$ Period of rotation $T = \frac{2\pi}{\omega \sin \lambda} = \frac{2\pi}{\omega \cdot \frac{1}{2}} = 24 \times 2 = 48$ hours.

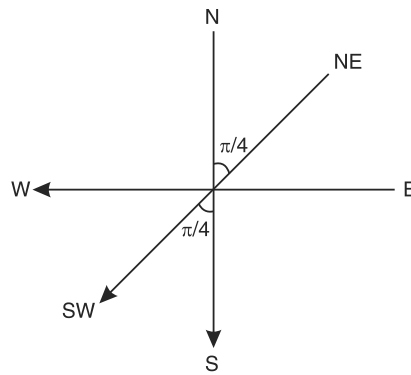


Fig. 2.17

Hence rate of rotation = $\frac{2\pi}{48}$ radian is per hour.

In turning from NS to NE-SW direction the pendulum turns through $45^\circ = \frac{\pi}{4}$ radian as shown.

$$\therefore \text{Time taken} = \frac{\frac{\pi}{4}}{2\pi/48} = \frac{48}{8} = 6 \text{ hours.}$$

Example 26. Show that at latitude θ , the plane of oscillation of Foucault's pendulum rotates through $2\pi \sin \theta$ every day. Explain this physically on a pole of the earth.

Solution. The period of rotation of a Foucault's pendulum at a latitude θ is given by

$$T = \frac{2\pi}{\omega \sin \theta}$$

where $\frac{2\pi}{\omega} = 24 \text{ hours.} \quad \therefore T = \frac{24}{\sin \theta} \text{ hours}$

In T hours the plane rotates through 2π radians.

$\therefore$ In 24 hours the plane will rotate through an angle

$$\phi = \frac{2\pi}{T} \times 24 = \frac{2\pi}{24} \sin \theta \times 24 = 2\pi \sin \theta \text{ radian}$$

At the poles $\phi = 90^\circ, \sin \theta = \sin 90^\circ = 1$

$\therefore \phi = 2\pi \text{ radians}$

Thus the plane of oscillation of a Foucault's pendulum rotates through 2π radians in 24 hours at the poles which means that the period of rotation of the earth about its own axis is 24 hours.

Example 27. Calculate the rate of rotation of the plane of oscillation of a pendulum in latitude 30° and hence obtain the time it will take to turn through a full right angle.

Solution. We know that the time taken by the plane of oscillation of a pendulum to make one complete rotation in latitude ϕ is given by $T = 24/\sin \phi$ hours.

Also, if α be the angular velocity of the plane of oscillation in *this* latitude, its time-period (*i.e.*, time taken for one full rotation) must be $T = 2\pi/\alpha$.

We, therefore, have $2\pi/\alpha = 24/\sin \phi$, whence, $\alpha = 2\pi \sin \phi/24$ radian/hr.

Or, rate of rotation of the plane of oscillation of the pendulum in latitude 30°
 $= 360^\circ \sin \phi/24 = 15^\circ \sin \phi$ per hour.

Since, here, $\phi = 30^\circ$, and therefore, $\sin \phi = \frac{1}{2}$, we have

$$\begin{aligned} \text{rate of rotation of the plane of oscillation of the pendulum in latitude } 30^\circ \\ = 15 \times \frac{1}{2} = 7.5^\circ/\text{hour.} \end{aligned}$$

$$\begin{aligned} \therefore \text{Time taken by the plane of oscillation to turn through one full right angle} \\ = 90/7.5 = 12 \text{ hours.} \end{aligned}$$

[Or, we could have said that if T be the time-period of the plane of oscillation in hours, the time taken to turn through 90° must be $T/4$ hours.

$$\begin{aligned} &= \frac{24}{\sin \phi} \times \frac{1}{4} = \frac{24}{\sin 30^\circ} \times \frac{1}{4} \\ &= \frac{24 \times 2}{4} = 12 \text{ hours.}] \end{aligned}$$

Example 28. A pendulum is oscillating along the east-west direction at a place of latitude 30°N . After how much time will it start oscillating along the north west-south east direction?

Solution. The problem here, obviously, is to calculate the time the plane of oscillation of the pendulum will take to turn through an angle of 135° , (Fig. 2.18) in latitude 30° N.

Since in latitude $\phi = 30^\circ$, the rate of rotation of the plane of oscillation of the pendulum is (as we have seen, under example 11 above) equal to $15^\circ \sin \phi = 15^\circ \sin 30^\circ = 7.5^\circ/\text{hr}$, we have, time taken by the plane of oscillation to turn through, $135^\circ = 135/7.5 = 18$ hours.

Thus, *the pendulum will start oscillating along the north west-south east direction after 18 hours.*

Example 29. A satellite moves in a circular orbit around the earth at a height $R_e/2$ from the earth's surface, where R_e is the radius of the earth. Calculate its period of revolution.

Given $R_e = 6.38 \times 10^6 \text{ m}$, $g = 9.80 \text{ m/s}^2$.

We have seen under article 2.14 that the time-period of a satellite revolving the earth at a height h from its surface is given by $T = 2\pi\sqrt{(R+h)^3/gR^2}$, where R is the radius of the earth.

Here, $R = R_e$ and $h = R_e/2$. So that, *time-period of revolution of the satellite,*

$$T = 2\pi\sqrt{\left(R_e + \frac{R_e}{2}\right)^3 / g(R_e)^2} = 2\pi\sqrt{27R_e/8g}.$$

Substituting the given values of R_e and g , therefore, *we have time period of the satellite,*

$$T = 2\pi\sqrt{\frac{27 \times 6.38 \times 10^4}{8 \times 9.80}} = 9.309 \times 10^3 \text{ or } 9.3 \times 10^3 \text{ s}$$

Example 30. At what height from the surface of the earth must a satellite revolve in its orbit around the earth, concentric and coplanar with the equator, such that it always appears to be at the same point over the earth's surface as viewed by an observer on the earth? ($G = 6.67 \times 10^{-8} \text{ c.g.s units}$, $R = 6.38 \times 10^8 \text{ cm}$ and mass of the earth $= 5.98 \times 10^{27} \text{ g}$).

Solution. Obviously, the satellite will always appear to be at the same point over the earth's surface when its angular velocity is the *same* as that of the earth,

viz., $\omega = 2\pi/24 \times 60 \times 60 = 7.27 \times 10^{-5} \text{ radians or } 7.3 \text{ radians.}$

Let r be the radius of the orbit of the satellite. Then, as we have seen under article 2.14, *angular velocity of the satellite, i.e.*, $\omega = \sqrt{MG/r^3}$. Or, $\omega^2 = MG/r^3$, whence, $r^3 = MG/\omega^2$. So that, substituting the values of M , G and ω , we have

$$r^3 = \frac{5.98 \times 10^{27} \times 6.67 \times 10^{-8}}{(7.3 \times 10^{-5})^2} = 74.85 \times 10^{27}, \text{ whence, } r = (74.85 \times 10^{27})^{\frac{1}{3}} = 4.2 \times 10^9 \text{ cm.}$$

Now, $r = (R + h)$, where R is the radius of the earth and h , the height of the satellite from the surface of the earth.

We, therefore, have $h = r - R = 4.2 \times 10^9 - 6.38 \times 10^8 = 3.562 \times 10^9 \text{ cm.}$

Thus, *the satellite must revolve in its orbit around the earth at a height of $3.56 \times 10^9 \text{ cm}$ from the surface of the earth.*

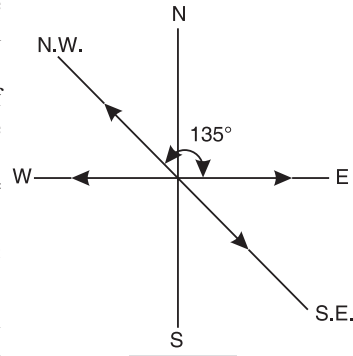


Fig. 2.18

EXERCISE

I. Inertial frames of reference—Galilean invariance

1. State Newton's laws of motion. Show that Newton's first law of motion is simply a special case of the second law. Discuss the limitations of Newton's laws of motion.

(Nagpur Uni. 2007, 2006, 2003)

2. (i) State and explain Newton's second law of motion.
 (ii) State the limitations of Newton's law of motion.
 (iii) Derive the equations for components of a velocity, Acceleration and force in a cartesian coordinate system.
3. A particle is moving along a curve in a plane. Derive expression for its radial and transverse components of velocity and acceleration.
4. A particle is moving along a curve in a plane. Derive expression for the radial and transverse component of velocity and acceleration. Prove that for the motion of a particle in a plane.

(Nagpur Uni. 2007)

(Nagpur Uni. 2007, 2006)

$$\vec{v} = r\dot{r}\hat{r} + r\dot{\theta}\hat{\theta} \text{ and } \vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

where the letters have their usual meaning

(Cal. Uni. 2003)

5. What is meant by a frame of reference? Give examples to show that the state of motion of a body would appear to be different when viewed from different frames of reference. From which particular frame of reference would you like to study this motion and why?
6. Show that to a person at rest on the ground, the path of a particle on the rim of a slowly moving wheel of a car will appear to be a cycloid while it will appear to be circular to an observer situated at the centre of the wheel.

[Hint. Imagine the motion of the wheel to be along the +x direction in the x-y plane, with OX along the horizontal and OY along the vertical, the origin O lying at the centre of the wheel. Let r be the radius of the wheel, $-\omega$, its angular velocity (being clockwise) and $-\theta$, the angle that a point P on its rim makes with the x-axis at O, to start with (Fig. 2.19). i.e., at $t = 0$. Then, after time t, the position of the point is specified by the coordinates $x = r \cos(-\omega t - \theta) = r \cos(\omega t + \theta)$, $y = -r \sin(\omega t + \theta)$, indicating that the point moves along a circular path of radius r, the same as that of the wheel.

For an observer, at rest on the ground, on the other hand, the centre of the wheel has also a linear velocity rw . Hence, after time t, the position of the point is given by the coordinates $x = r\omega t + r \cos(\omega t + \theta)$, $y = r - r \sin(\omega t + \theta)$, and this is the equation of a cycloid.]

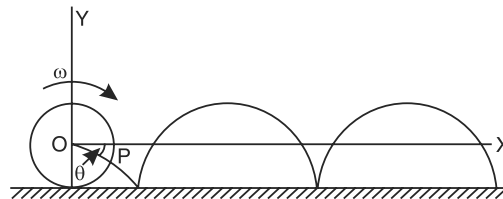


Fig. 2.19

7. Enunciate Newton's laws of motion and discuss their limitations.
8. What is an *inertial frame of reference*? Show that all other frames of reference, with constant velocity relative to it, are also inertial frames. What are the characteristic properties and importance of such frames? Can the earth be regarded as one such frame?
9. What is meant by *Galilean transformation* and *Galilean invariance*? Show that whereas length (or distance) and acceleration are invariant to Galilean transformation, velocity is not.
10. A reference frame S' is moving with a constant velocity v with respect to an inertial frame S and the position of its origin O' is given by R_0 at $t = 0$. If the position vector of a point in frame S be r , show that it will be $i + vt + R_0$ in frame S' .
11. Explain how the parabolic trajectory of a projectile would appear when observed from a frame of reference moving with (i) a velocity equal to the horizontal component of the velocity of the projectile, (ii) a velocity equal to twice the horizontal component of the velocity of the projectile, and (iii) a constant velocity inclined to the horizontal.

Ans. (i) a vertical straight line, (ii) original trajectory in the opposite direction (iii) a new parabolic trajectory.

12. Two inertial frames S and S' have their axes parallel and the position of the origin O' of frame S' relative to origin O of frame S is given by $\vec{r}_0 = \hat{i} + 2\hat{j} + 3\hat{k}$. Show that, if the position of a point P in frame S be $(2, 3, 4)$, its position if frame S' would be $(\hat{i} + \hat{j} + \hat{k})$.
13. If the respective axes of two frames S and S' be inclined to each other, with their origins coinciding, obtain the usual transformation equations and show that if frame S be inertial, frame S' too is inertial.
14. If in question 13 above, the position vector of the origin O' of frame S' with respect to the origin O of frame S be R_0 , show that $x = x_0 + x'C'_{xx} + y'C'_{yx} + z'C'_{zx}$; $y = y_0 + x'C'_{xy} + y'C'_{yy} + z'C'_{zy}$ and $z = z_0 + x'C'_{xz} + y'C'_{yz} + z'C'_{zz}$.
15. Enunciate the laws of conservation of momentum and conservation of energy and show that they are both invariant to Galilean transformation.
16. A particle of mass m_1 moving with velocity v collides *head on* against a particle of mass m_2 at rest. Taking the collision to be an *elastic* one obtain the velocities of the two particles and the fraction of the total kinetic energy acquired by the second particle after the collision. How will this fraction change if $m_1 = m_2$?

[Ans. $v_1 = \frac{m_1 v - m_2 v_2}{m_1}$; $v_2 = \frac{2v}{1 + m_1/m_2}$; $\frac{4m_1 m_2}{(m_1 + m_2)}$; the fraction will change to 1, i.e., the entire energy of m_1 will be transferred to m_2 .]

17. In question 16 above, if the collision be an *inelastic* one, so that the two particles coalesce together on collision, obtain their common velocity after the collision and the loss of kinetic energy suffered.

[Ans. Common velocity = $\frac{m_1 v}{m_1 + m_2}$; loss of energy suffered = $\frac{1}{2} m_1 v \frac{m_2}{m_1 + m_2}$.]

18. In a *head-on* collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. Find the velocities v_1 and v_2 of the two particles respectively after the collision if (a) half the original kinetic energy is lost, (b) the final kinetic energy is 50% greater than the original kinetic energy.

[Ans. (a) $v_1 = v_2 = \frac{1}{2} v_0$, (b) $v_1 = \frac{1}{2} v_0(1 + \sqrt{2})$, $v_2 = \frac{1}{2} v_0(1 - \sqrt{2})$.]

19. The position vectors of two particles are respectively $\vec{r}_1 = 4\hat{i} + 6\hat{j} + 4\hat{k}$ m. and $\vec{r}_2 = 2\hat{i} + 3\hat{j} + 2\hat{k}$ m. The velocity of the first particle is $\hat{i} - 2\hat{j} + 2\hat{k}$ m/s. How much greater should the velocity of the second particle be so that the two may collide in 5 seconds? [Ans. $0.2(2\hat{i} + 3\hat{j} + 2\hat{k})$ m/s.]

II. Non-inertial frames—Fictitious forces

20. What is meant by a non-inertial frame of reference? And what is a fictitious force? Why? is it so called? Under what condition will an accelerated frame of reference serve as an inertial frame?
21. What will be the effective weight of a person carried vertically up in a rocket with an acceleration of $6g$, if his actual weight on the earth is 60 kg. [Ans. 420 kg.]
22. A car carries a framework $ABCD$, shown in Fig. 2.20 in which a mass P of 200 g is supported between two springs of force constant 5000 dyne/cm each. Side AB is kept horizontal and is along the length of the car and the pointer attached to P reads zero when the car is at rest. What will the pointer read when the car has (i) a uniform speed of 20 m/s on a straight road, (ii) a uniform speed of 10 m/s on a circular road of radius 20 metres,

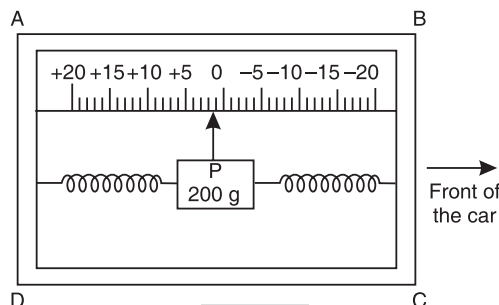


Fig. 2.20

(iii) uniform acceleration of 0.5 m/s on a straight road, (iv) a uniform acceleration of -1.0 m/s on a straight road ?
[Ans. (i) 0, (ii) 0, (iii) $+10$ mm (iv) -20 mm.]

[Hint. (i) No acceleration of the mass relative to the car. Hence no fictitious force.
 (ii) Centripetal acceleration towards the centre of the circular path—hence a fictitious force, viz., *centrifugal force* perpendicular to AB, *outwards*, counter balancing the centripetal force (inwards)—no change in the position of the pointer which, therefore, continues to read zero. (iii) Acceleration *forwards*—hence fictitious force $= ma$. (iv) Here, since a is negative the fictitious force $= +ma$.]

23. A person weighs 50 kg on the earth. Calculate the fictitious force acting on him and his effective weight in a lift when the lift is moving (i) down, (ii) up, with an acceleration of 4m/s.

[Ans. (i) 200 N, upwards; 29.6 kg, (iii) 200 N, downwards; 70.4 kg.]

24. What is *Coriolis force*? Under what conditions does it come into play? Discuss in general terms the effect of the Coriolis force produced as a result of the earth's rotation. What is *Ferrel's law*?

25. Calculate the values of the centrifugal and the Coriolis forces on a mass of 20 g placed at a distance of 10 cm from the axis of a rotating frame of reference, if the angular speed of rotation of the frame be 10 radians per second.
[Ans. 2×10^4 dyne; 4×10^4 dyne.]

26. A mass of ice, weighing 500,000 tons near the North pole moves west at the rate of 5 miles a day. Neglecting the curvature of the earth, find the magnitude and direction of the Coriolis force. Express the magnitude in tons weight (Take $g = 32$ m/s).
[Ans. 0.7 tons weight, due north.]

27. A reference frame a rotates with respect to another reference frame b with uniform angular velocity ω . If the position, velocity and acceleration of a particle in frame a are represented by R , v_a and f_a respectively, show that the acceleration of that particle in frame b is given by $f_b = f_a + 2\omega \times v_a + \omega \times (\omega \times R)$. Interpret this equation with reference to the motion of bodies on the earth's surface.

28. Obtain an expression for the horizontal displacement of an object falling freely under the action of gravity in latitude ϕ . Hence calculate the displacement suffered by a stone dropped from the top of a tower 100 metres high (i) in latitude 60°N , (ii) at the equator. (Angular velocity of the earth about its axis $= 7.2 \times 10^{-5}$ m/s and $g = 9.8$ m/s²).
[Ans. (i) 1.085 cm; (ii) 2.17 cm.]

29. Calculate the fictitious acceleration of the sun relative to a reference frame fixed on the surface of the earth. (Distance between the sun and the earth $= 1.5 \times 10^{12}$ cm).

[Ans. 7.8×10^4 cm/s², directed towards the earth.]

III. Foucault's pendulum—Earth satellites

30. What is a Foucault pendulum? How does it enable us to demonstrate the rotation of the earth about its own axis?

31. What will be the period of rotation of the plane of oscillation of Foucault's pendulum (i) in latitudes 30° and 45° , (ii) at the North pole, (iii) at the equator?

[Ans. (i) 12 hours and 33.94 hours (ii) 24 hours, (iii) Infinite.]

32. A pendulum is oscillating along the N-S direction at a place in latitude 30° N. How long will it take to start oscillating along N.E.–S.W. direction?
[Ans. 6 hours.]

33. If the rotation of the earth about its own axis were to cease suddenly, how will it effect the value of g at place in latitude 45° ? (Radius of the earth $= 6.4 \times 10^8$ cm)

[Ans. Increase of 1.69 cm/s² in the value of g .]

[Hint. We know that $g\phi = g.R\omega^2 \cos^2\phi$. If $\omega = 0$, the value of $g\phi$ becomes g , i.e., increases by $g - g\phi = g - (g - R\omega^2 \cos^2\phi) = R\omega^2 \cos^2\phi = R\omega^2 \cos^2 45^\circ = R\omega^2/2$, where, $R = 6.4 \times 10^8$ cm and $\omega = 2\pi/24 \times 60 \times 60$ radian/s.]

34. Assuming the law of gravitation, find an expression for the period of revolution of a satellite.

Calculate the mass of the earth and the time of revolution of a satellite describing a circular orbit close to the earth's surface from the following data: $G = 6.6 \times 10^{-8}$ c.g.s. units, $g = 981$ cm./s² and *radius of the earth*, $R = 6.37 \times 10^8$ cm.

[Ans. 6.30×10^{27} cm.; 5060 s².]

35. Considering the earth to be homogeneous sphere of mass density σ , show that the period of revolution of a satellite in an orbit lying just outside the equator depends only upon the density σ .

[Hint. We know that the period of revolution of a satellite is given by $T = 2\pi\sqrt{(R+h)^3/gR^2}$ Putting $gR^2 = MG$, $T = 2\pi\sqrt{(R+h)^3/MG}$, where M is mass of the earth and R its radius. Since the satellite is close to earth's surface $(R+h) \cong R$. Hence $T = 2\pi\sqrt{R^3/MG}$ Clearly $M = \frac{4}{3}\pi R^3 \sigma$.

Therefore, $T = 2\pi\sqrt{3R^3/4\pi R^3 G \sigma} = \sqrt{3\pi/\sigma G}$, showing that T depends only upon σ , G being universal constant].